A

PROJECT REPORT

ON

"SONREB – COMBIND NON DESTRUCTIVE TESTING"

In the partial fulfillment of the requirement for Bachelor Degree in Civil Engineering

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This work is being submitted for the award of degree of Bachelor of Civil Engineering .It is submitted in the partial fulfilment of the prescribed syllabus of Savitribai Phule University of Pune, for academic year 2014-15.

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SYNOPSIS

The attention paid to the ageing of civil engineering structures and the urgent need for their rehabilitation is relatively recent. The rehabilitation of structures clearly revealed the limit of knowledge in the field of the evaluation to gather the data necessary to the analyses. The present need, which will develop further in the next years, is directed towards the Non-destructive testing and instrumentation of structures to obtain in-situ characteristics and properties of construction materials.

Initially, Non-destructive testing techniques were introduced in civil engineering during the 40's. The principal need of the engineers was the in situ determination of the homogeneity and the compressive strength of fresh concrete to be able, for example, to remove the formworks. The majority of these techniques (rebound hammer, pull-out test, Windsor probe test, etc.) are standardized and are based on the measurement of the surface hardness of the concrete. With the progressive ageing of the structures, the needs of the engineers evolved to the search for tools allowing the estimation of the mechanical properties of old materials, as well as the detection and the characterization of hidden defects.

In this paper a series of non-destructive tests has been performed with the purpose to investigate on the mechanical properties of the concrete employed in the civil buildings. A series of specimens were prepared in order to correlate the “in situ” concrete strengths obtained by multiple combined non-destructive methods with the cubical strength obtained by destructive methods. The comparison among non-destructive tests and cubical compression strength are carried out.

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CHAPTER - 1

INTRODUCTION

1.1 General Introduction

The evaluation of the concrete compressive strength is a fundamental step for assessment of existing reinforced concrete building according to the last seismic codes. This valuation can be conducted by the use of both destructive and non-destructive method, although being minimally invasive and easily extensible to a large number of elements, are influenced by many factors.

Concrete is a variety of environmental degrading factors which tend to limit its service life. This therefore has brought about the need for test method to measure the in-place properties of concrete for quality assurance and for evaluation of existing conditions. Since such test are expected not to impair the function of the structure and allow for re-testing at the same location to evaluate the changes in property at some other point in time, these methods should be non-destructive. Non-destructive test as applied to concrete are those that do not alter the concrete quality. Non-destructive Testing (NDT) as the name implies refer to a test that does not impair the intended performance of the element, member or structure under investigation.

At present the test use mainly as a basis of quality control is compression testing of cubes and it represents the potential strength of the concrete used. The main parameters determining the quality of concrete is its composition, compaction and curing. At the most it can be ensured that the composition of concrete going into the cubes and that going into the structure is the same. However, the method of compaction and curing usually are different for the cubes and the structural members and as such the results obtained on cubes may not truly represent the quality of concrete in the structure, Hence the need of Non-destructive Testing (NDT).

NDT of concrete is of great scientific and practical importance especially the need for quality characterization of damaged constructions made of concrete. Its importance can be seen in the desire for a proposed change of usage or extension of a structure, acceptability of a structure for purchase or insurance, assessment of the

quality or integrity of apair, monitoringof strength, curing, pre-casting or load application.

This research therefore seeks to compare the most common non-destructive techniques, the re-bound hammer and the ultrasonic pulse velocity methods so as to see which method has a superior capability in the sense that it is capable of providing more information on concrete properties. This will aid in quality control on site as well as help in monitoring concrete strength development as so to hasten construction rate, facilitate possible correction of defects in structures, predict future performance and allow minor repairs.

The combination of several techniques of non-destructive testing is often implemented empirically, combining two techniques most often used to enhance the reliability of the estimate compressive strength of concrete; the principle is based on correlations between observed measurements and the desired property. The compressive strength of a concrete is usually the most sought after property. This is leads to the development of a method that combines index rebound hammer and the ultrasonic pulse velocity. The objective of the combined test is to evaluate the compressive strength of concrete in situ; the best approach is generally to develop a relationship of correlation between the UPV/the index of rebound hammer and the compressive strength of standardized laboratory specimen, in some cases specimen are not available, then a number of core must be taken to establish this relationship. The standardized combine method the most widely used in SonReb method. First born and established in Romania then developed in Australia and in Europe.

It is often necessary to test concrete structures after the concrete has hardened to determine whether the structure is suitable for its designed use. Ideally, such testing should be done without damaging the concrete. The tests available for testing concrete range from completely non-destructive tests, where there is no damage to the concrete, through those where the concrete surface is slightly damaged, to partially destructive tests, such as core tests and pull-out andpull-off tests, where the surface has to be repaired after the test. The range of properties that can be assessed using non-destructive tests and partially destructive tests is quite large and includes such fundamental parameters as density, elastic modulus and strength as well as surface hardness, surface absorption, reinforcement location, size and distance from the

surface. The crushing of the specimens is the usual destructive test to assess the strength of concrete. Non- destructive methods like rebound hammer test and ultrasonic test do not damage buildings and allow having an inventory of structures and conditions. Nondestructive tests are widely applied to study mechanical properties and integrity of concrete structures. They are simple to use and often economically advantageous. They are suitable for taking measurements on site and taking continuous measurements. These non-destructive methods are usually associated with each other to improve diagnosis and reduce the number of tests. Ultrasound measurements provide a simple non- destructive and inexpensive method to evaluate the elastic modulus of concrete. The formulae proposed by different standards to estimate the dynamic modulus of elasticity from the resistance are very approximate. The dynamic modulus of elasticity is strongly influenced by the aggregates; it cannot be determined accurately based on the strength, which depends mainly on the cement paste and the particle size.

This report presents measurements of compressive strength determined from non-destructive tests. The results obtained from non-destructive tests were compared with destructive test results. The influences of the age of the concrete, its strength and water/cement ratio on the resistance determined by rebound hammer test and compression test were studied. A simplified expression has been proposed to estimate the rebound number from the value of dynamic modulus of elasticity determined by pulse velocity test.

1.2 Purpose of Testing

A variety of methods has been developed or is under developments for investigating and evaluating concrete structures. These methods are aimed at estimation of strength and other properties; monitoring and assessing corrosion; measuring crack size and cover; assessing grout quality; detecting defects and identifying relatively more vulnerable areas in concrete structures.

Many of the methods used for concrete testing have their origin to the testing of more homogenous, metallic system. These methods have a sound scientific basis, but heterogeneity of concrete makes interpretation of results somewhat difficult. There could be many parameters such as material, mix, workmanship, environment, which influences the result of measurement. Moreover, this tests measure some other

property of concrete (e.g. hardness) and results are interpreted to assess a different property of concrete e.g. strength which is of primary interest.

Thus, interpretation of results is very important and difficult job where generalization is not possible. As such, operators can carry out tests but interpretation of results must be left to experts having experience and knowledge of application of such non-destructive tests.

It is often necessary to test concrete structures after the concrete has hardenedto determine whether the structure is suitable for its design use. Ideally such testing should be done without damaging the concrete. The test available for testing concrete range from the completely non-destructive, where there is no damage to the concrete through those where the concrete surface is slightly damaged, to partially destructive tests, such as core test and pull out and pull off test, where the surface has to repaired after the test.

Testing methods can be applied to both old and new structures. For new structures, the principal applications are likely to be for quality control or the resolution of doubts about the quality of material or construction. The testing of existing structures is usually related to an assessment of structural integrity or adequacy.

- **Purposes:-**
  - (i) Estimating the in-situ Compressive strength.
  - (ii) Estimating the uniformity and homogeneity.
  - (iii) Estimating the quality in relation to standard requirement.
  - (iv) Detection of presence of cracks, voids and other imperfections.
  - (v) Identifying areas of lower integrity in comparison to other parts.
  - (vi) Monitoring changes in the structure of the concrete which may occur with time.
  - (vii) Identifying of reinforcement profile and measurement of cover, bar diameter etc.
  - (viii) Condition of pre-stressing/reinforcement steel with respect to corrosion.
  - (ix) Chloride, Sulphate, alkali contents or degree of carbonation.
  - (x) Measurement of elastic modulus.
  - (xi) Condition of grouting in pre-stressing cable ducts.

1.3 Destructive Testing

As the name suggests, destructive testing (DT) includes methods where your material is broken down in order to determine mechanical properties, such as strength, toughness and hardness. In practice it means, for example, finding out if the quality of a weld is good enough to withstand extreme pressure or to verify the properties of a material.

These properties can't be examined with non-destructive methods, as specimens of the material must be extracted. Destructive testing is generally most suitable and economic for mass produced objects, as the cost of destroying a small number of pieces is negligible. The samples are put under different loads and stress. That way we can analyze in which point your material eventually gives up and cracks. The results gained are then compared to regulations and/or quality guidelines.

Destructive tests are best when used together with our non-destructive methods: this combination gives the best information on materials and welds. Non-destructive tests show if cracks, corrosion or other faults exist. Destructive tests in turn indicate how and when the objects are in danger of breaking down or failing.

Destructive testing services include mechanical testing (tensile, bend and impact tests), hardness testing, macro and micro testing as well as material analysis and metallographic examinations. Inspecta has its own, accredited destructive testing laboratories in several countries. We are also able to execute your required tests at our own workshops. This way we can ensure you get the best service and the most suitable testing methods.

- **Benefits of Destructive Testing (DT)**
  - Verifies properties of a material
  - Determines quality of welds
  - Helps you to reduce failures, accidents and costs
  - Ensures compliance with regulations

1.4 NON DESTRUCTIVE TESTING

Nondestructive testing (NDT) is the process of inspecting, testing, or evaluating materials, components or assemblies for discontinuities, or differences in

characteristics without destroying the serviceability of the part or system. In other words, when the inspection or test is completed the part can still be used. In contrast to NDT, other tests are destructive in nature and are therefore done on a limited number of samples ("lot sampling"), rather than on the materials, components or assemblies actually being put into service.

These destructive tests are often used to determine the physical properties of materials such as impact resistance, ductility, yield and ultimate tensile strength, fracture toughness and fatigue strength, but discontinuities and differences in material characteristics are more effectively found by NDT.

Today modern nondestructive tests are used in manufacturing, fabrication and in-service inspections to ensure product integrity and reliability, to control manufacturing processes, lower production costs and to maintain a uniform quality level. During construction, NDT is used to ensure the quality of materials and joining processes during the fabrication and erection phases, and in-service NDT inspections are used to ensure that the products in use continue to have the integrity necessary to ensure their usefulness and the safety of the public.

It should be noted that while the medical field uses many of the same processes, the term "nondestructive testing" is generally not used to describe medical applications**.**

Typical situations where non-destructive testing may be useful are, as follows:

- Quality control of pre-cast units or construction in situ
- Removing uncertainties about the acceptability of the material supplied owing to apparent non-compliance with specification
- Confirming or negating doubt concerning the workmanship involved in batching, mixing, placing, compacting or curing of concrete
- Monitoring of strength development in relation to formwork removal, cessation of curing, pre-stressing, load application or similar purpose
- Location and determination of the extent of cracks, voids, honeycombing and similar defects within a concrete structure
- Determining the concrete uniformity, possibly preliminary to core cutting, load testing or other more expensive or disruptive tests

- Determining the position, quantity or condition of reinforcement
- Increasing the confidence level of a smaller number of destructive tests
- Determining the extent of concrete variability in order to help in the selection of sample locations representative of the quality to be assessed
- Confirming or locating suspected deterioration of concrete resulting from such factors as overloading, fatigue, external or internal chemical attack or change, fire, explosion, environmental effects
- Assessing the potential durability of the concrete
- Monitoring long term changes in concrete properties
- Providing information for any proposed change of use of a structure for insurance or for change of ownership.

➢ **Basic methods for NDT of concrete structures**

The following methods, with some typical applications, have been used for the NDT of concrete:

- **Visual testing**, which is an essential precursor to any intended non-destructive test. An experienced civil or structural engineer may be able to establish the possible cause(s) of damage to a concrete structure and hence identify which of the various NDT methods available could be most useful for any further investigation of the problem.
- **Half-cell electrical potential method**, used to detect the corrosion potential of reinforcing bars in concrete.
- **Schmidt/rebound hammer test**, used to evaluate the surface hardness of concrete.
- **Carbonation depth measurement test**, used to determine whether moisture has reached the depth of the reinforcing bars and hence corrosion may be occurring.
- **Permeability test**, used to measure the flow of water through the concrete.
- **Penetration resistance or Windsor probe test**, used to measure the surface hardness and hence the strength of the surface and near surface layers of the concrete.

- **Cover meter testing**, used to measure the distance of steel reinforcing bars beneath the surface of the concrete and also possibly to measure the diameter of the reinforcing bars.
- **Radiographic testing,** used to detect voids in the concrete and the position of stressing ducts.
- **Ultrasonic pulse velocity testing**, mainly used to measure the sound velocity of the concrete and hence the compressive strength of the concrete.
- **Sonic methods** using an instrumented hammer providing both sonic echo and transmission methods.
- **Tomographic modeling**, which uses the data from ultrasonic transmission tests in two or more directions to detect voids in concrete.
- **Impact echo testing**, used to detect voids, de-lamination and other anomalies in concrete.
- **Ground penetrating radar or impulse radar testing**, used to detect the position of reinforcing bars or stressing ducts.
- **Infrared thermography**, used to detect voids, de-lamination and other anomalies in concrete and also detect water entry points in buildings.

1.4.1 Schmidt Rebound Hammer Test

1.4.1.1 Fundamental Principle

The Schmidt rebound hammer is principally a surface hardness tester. It works on the principle that the rebound of an elastic mass depends on the hardness of the surface against which the mass impinges. There is little apparent theoretical relationship between the strength of concrete and the rebound number of the hammer. However, within limits, empirical correlations have been established between strength properties and the rebound number. Further, Kolek has attempted to establish a correlation between the hammer rebound number and the hardness as measured by the Brinell method.

1.4.1.2 EQUIPMENT FOR SCHMIDT/REBOUND HAMMER TEST

The Schmidt rebound hammer is shown in Fig. 1.1. The hammer weighs about 1.8 kg and is suitable for use both in a laboratory and in the field. A schematic cutaway view of the rebound hammer is shown in Fig. 1.2. The main components

include the outer body, the plunger, the hammer mass, and the main spring. Other features include a latching mechanism that locks the hammer mass to the plunger rod and a sliding rider to measure the rebound of the hammer mass. The rebound distance is measured on an arbitrary scale marked from 10 to 100. The rebound distance is recorded as a "rebound number" corresponding to the position of the rider on the scale.



Figure 1.1 Schmidt Rebound Hammer

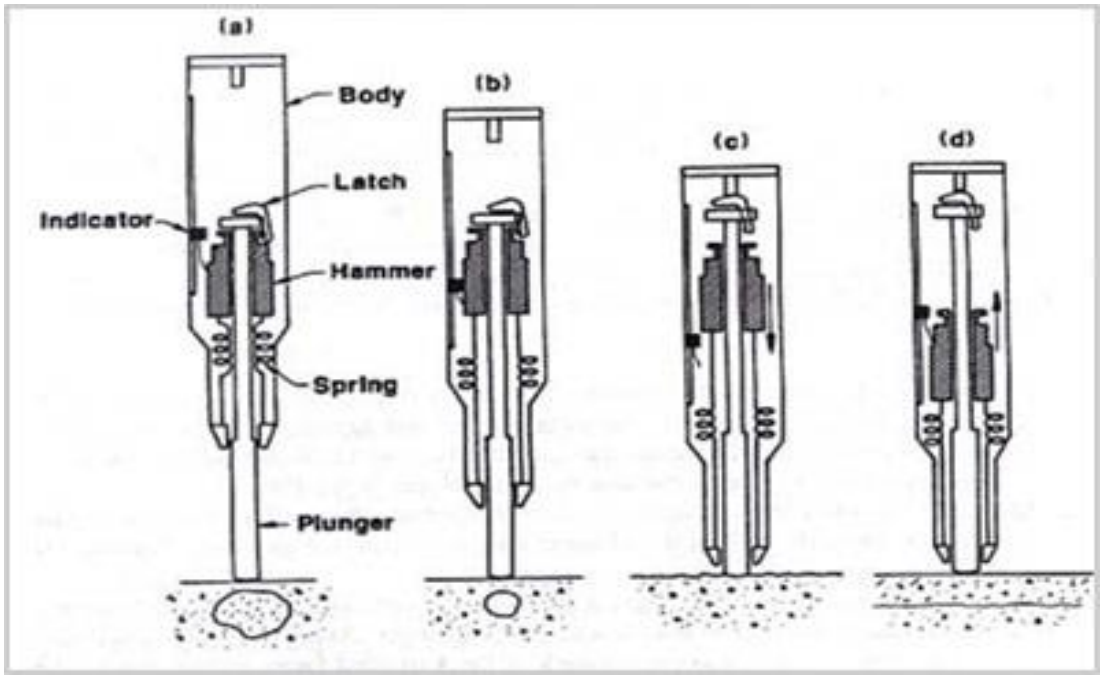


Figure 1.2 Cutaway View of the Rebound Hammer

- **General Procedure for Schmidt Rebound Hammer Test**

The method of using the hammer is explained using Fig. 1.2. With the hammer pushed hard against the concrete, the body is allowed to move away from the concrete until the latch connects the hammer mass to the plunger, Fig. 1.2(a).

The plunger is then held perpendicular to the concrete surface and the body pushed towards the concrete, Fig. 1.2(b). This movement extends the spring holding the mass to the body. When the maximum extension of the spring is reached, the latch releases and the mass is pulled towards the surface by the spring, Fig. 1.2(c).The mass hits the shoulder of the plunger rod and rebounds because the rod is pushed hard against the concrete, Fig. 1.2(d). During rebound the slide indicator travels with the hammer mass and stops at the maximum distance the mass reaches after rebounding. A button on the side of the body is pushed to lock the plunger into the retracted position and the rebound number is read from a scale on the body.

- **Applications of Schmidt Rebound Hammer Test**

The hammer can be used in the horizontal, vertically overhead or vertically downward positions as well as at any intermediate angle, provided the hammer is perpendicular to the surface under test. The position of the mass relative to the vertical, however, affects the rebound number due to the action of gravity on the mass in the hammer. Thus the rebound number of a floor would be expected to be smaller than that of a soffit and inclined and vertical surfaces would yield intermediate results. Although a high rebound number represents concrete with a higher compressive strength than concrete with a low rebound number, the test is only useful if a correlation can be developed between the rebound number and concrete made with the same coarse aggregate as that being tested. Too much reliance should not be placed on the calibration curve supplied with the hammer since the manufacturer develops this curve using standard cube specimens and the mix used could be very different from the one being tested.

- **Range and Limitations of Schmidt Rebound Hammer Test**

Although the rebound hammer does provide a quick, inexpensive method of checking the uniformity of concrete, it has some serious limitations. The results are affected by:

1. **Smoothness of the test surface**

   Hammer has to be used against a smooth surface, preferably a formed one. Open textured concrete cannot therefore be tested. If the

surface is rough, e.g. a trowelled surface, it should be rubbed smooth with a carborundum stone.

2. Size, shape and rigidity of the specimen

If the concrete does not form part of a large mass any movement caused by the impact of the hammer will result in a reduction in the rebound number. In such cases the member has to be rigidly held or backed up by a heavy mass.

3. Age of the specimen

For equal strengths, higher rebound numbers are obtained with a 7 day old concrete than with a 28 day old. Therefore, when old concrete is to be tested in a structure a direct correlation is necessary between the rebound numbers and compressive strengths of cores taken from the structure. Rebound testing should not be carried out on low strength concrete at early ages or when the concrete strength is less than 7 MPa since the concrete surface could be damaged by the hammer.

4. Surface and internal moisture conditions of concrete

The rebound numbers are lower for well-cured air dried specimens than for the same specimens tested after being soaked in water and tested in the saturated surface dried conditions. Therefore, whenever the actual moisture condition of the field concrete or specimen is unknown, the surface should be pre-saturated for several hours before testing. A correlation curve for tests performed on saturated surface dried specimens should then be used to estimate the compressive strength.

5. Type of coarse aggregate

Even though the same aggregate type is used in the concrete mix, the correlation curves can be different if the source of the aggregate is different. An example is shown in Fig. 4.5 where correlation curves for four different sources of gravel are plotted.

6. Type of cement

High alumina cement can have a compressive strength 100% higher than the strength estimated using a correlation curve based on ordinary Portland cement. Also, super sulphated cement concrete can have strength 50% lower than ordinary Portland cement.

7. Carbonation of the concrete surface

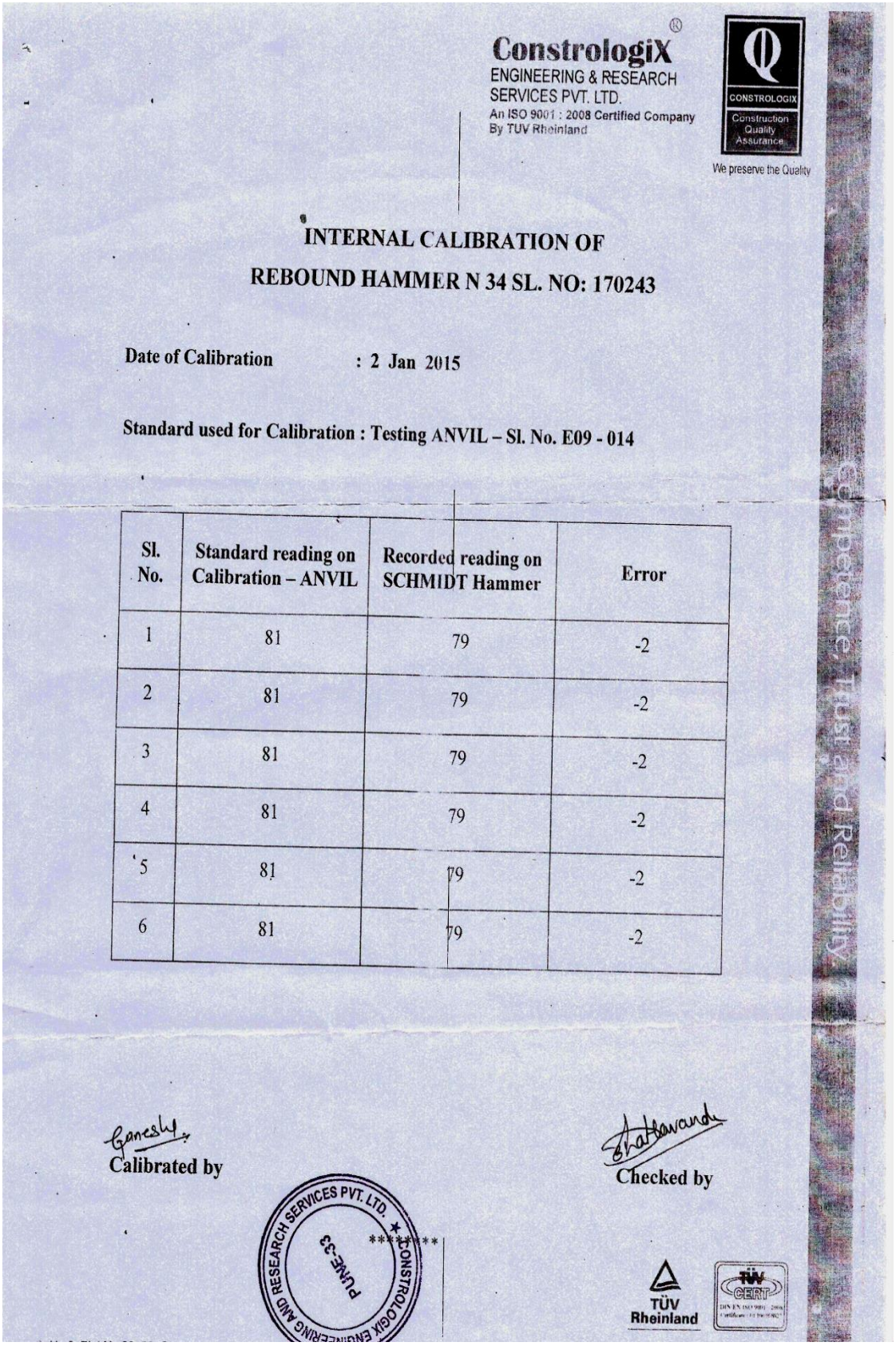
In older concrete the carbonation depth can be several millimeters thick and, in extreme cases, up to 20 mm thick. In such cases the rebound numbers can be up to 50% higher than those obtained on an uncarbonated concrete surface.

- **Calibration**

In this method of determining the compressive strength of concrete by rebound hammer, the instrument is first calibrated using a basic tool called ‘Anvil’. Anvil has a hard surface on which another instrument is struck. It has a Rebound Number of 81.



Figure 1.3 an Anvil



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INTERNAL CALIBRATION OF

REBOUND HAMMER N 34 SL. NO: 170243

Date of Calibration : 2 Jan 2015

Standard used for Calibration : Testing ANVIL – Sl. No. E09 - 014

Sl. No.	Standard reading on Calibration – ANVIL	Recorded reading on SCHMIDT Hammer	Error
1	81	79	-2
2	81	79	-2
3	81	79	-2
4	81	79	-2
5	81	79	-2
6	81	79	-2

Calibrated by

Checked by

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Figure 1.4 Rebound Hammer Calibration Certificate

1.4.2 ULTRASONIC TESTING

1.4.2.1 FUNDAMENTAL PRINCIPLE

A pulse of longitudinal vibrations is produced by an electro-acoustical transducer, which is held in contact with one surface of the concrete under test. When the pulse generated is transmitted into the concrete from the transducer using a liquid coupling material such as grease or cellulose paste, it undergoes multiple reflections at the boundaries of the different material phases within the concrete. A complex system of stress waves develops, which include both longitudinal and shear waves, and propagates through the concrete. The first waves to reach the receiving transducer are the longitudinal waves, which are converted into an electrical signal by a second transducer. Electronic timing circuits enable the transit time T of the pulse to be measured.

Longitudinal pulse velocity (in km/s or m/s) is given by:

$$UPV = \frac{L}{T} \quad \text{(m/s)}$$

Where, L is the path length,
T is the time taken by the pulse to traverse that length.

1.4.2.2 EQUIPMENT FOR ULTRASONIC PULSE VELOCITY

The equipment consists essentially of an electrical pulse generator, a pair of transducers, an amplifier and an electronic timing device for measuring the time interval between the initiation of a pulse generated at the transmitting transducer and its arrival at the receiving transducer. Two forms of electronic timing apparatus and display are available, one of which uses a cathode ray tube on which the received pulse is displayed in relation to a suitable time scale, the other uses an interval timer with a direct reading digital display. The equipment should have the following characteristics. It should be capable of measuring transit time over path lengths ranging from about 100 mm to the maximum thickness to be inspected to an accuracy of ±1%. Generally the transducers used should be in the range of 20 to 150 kHz although frequencies as low as 10 kHz may be used for very long concrete path lengths and as high as 1 MHz for mortars and grouts or for short path lengths. High

frequency pulses have a well defined onset but, as they pass through the concrete, become attenuated more rapidly than pulses of lower frequency. It is therefore preferable to use high frequency transducers for short path lengths and low frequency transducers for long path lengths. Transducers with a frequency of 50 kHz to 60 kHz are suitable for most common applications.

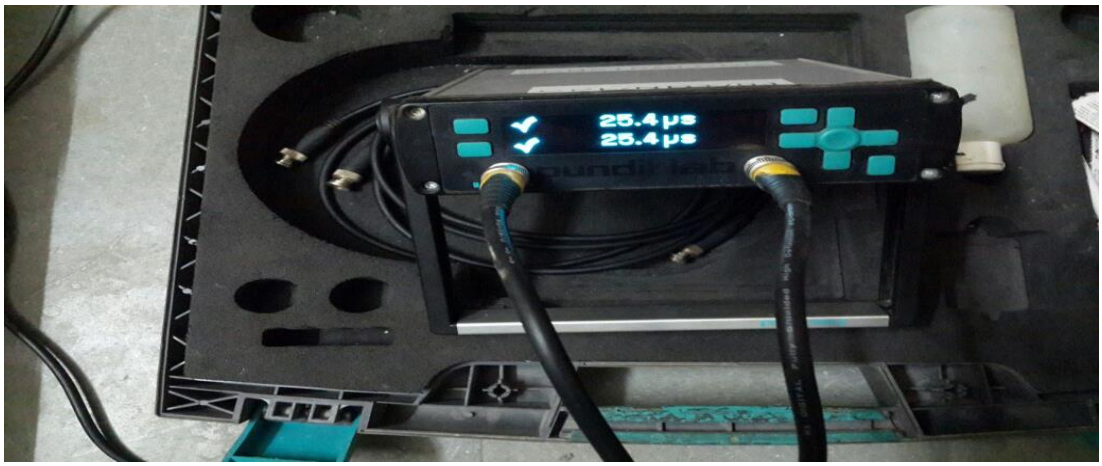


Figure 1.5 UPV Tester

Applicationsof Ultrasonic Pulse Velocity

Measurement of the velocity of ultrasonic pulses of longitudinal vibrations passing through concrete may be used for the following applications:

- Determination of the uniformity of concrete in and between members
- Measurement of changes occurring with time in the properties of concrete
- Correlation of pulse velocity and strength as a measure of concrete quality.
- Determination of the modulus of elasticity and dynamic Poisson's ratio of the concrete.

The velocity of an ultrasonic pulse is influenced by those properties of concrete which determine its elastic stiffness and mechanical strength. The variations obtained in a set of pulse velocity measurements made along different paths in a structure reflect a corresponding variation in the state of the concrete. When a region of low compaction, voids or damaged material is present in the concrete under test, a corresponding reduction in the calculated pulse velocity occurs and this enables the approximate extent of the imperfections to be determined. As concrete matures or

deteriorates, the changes, which occur with time in its structure, are reflected in either an increase or a decrease, respectively, in the pulse velocity. This enables changes to be monitored by making tests at appropriate intervals of time. Pulse velocity measurements made on concrete structures may be used for quality control purposes. In comparison with mechanical tests on control samples such as cubes or cylinders, pulse velocity measurements have the advantage that they relate directly to the concrete in the structure rather than to samples, which may not be always truly representative of the concrete in situ. Ideally, pulse velocity should be related to the results of tests on structural components and, if a correlation can be established with the strength or other required properties of these components, it is desirable to make use of it. Such correlations can often be readily established directly for pre-cast units and can also be found for in situ work. Empirical relationships may be established between the pulse velocity and both the dynamic and static elastic moduli and the strength of concrete. The latter relationship is influenced by a number of factors including the type of cement, cement content, admixtures, type and size of the aggregate, curing conditions and age of concrete. Caution should be exercised when attempting to express the results of pulse velocity tests in terms of strengths or elastic properties, especially at strengths exceeding 60 MPa.

- **Determination of Pulse Velocity**

1. Transducer arrangement

The receiving transducer detects the arrival of that component of the pulse, which arrives earliest. This is generally the leading edge of the longitudinal vibration. Although the direction in which the maximum energy is propagated is at right angles to the face of the transmitting transducer, it is possible to detect pulses, which have travelled through the concrete in some other direction. It is possible, therefore, to make measurements of pulse velocity by placing the two transducers on either:

- Opposite faces (direct transmission)
- Adjacent faces (semi-direct transmission): or
- The same face (indirect or surface transmission).

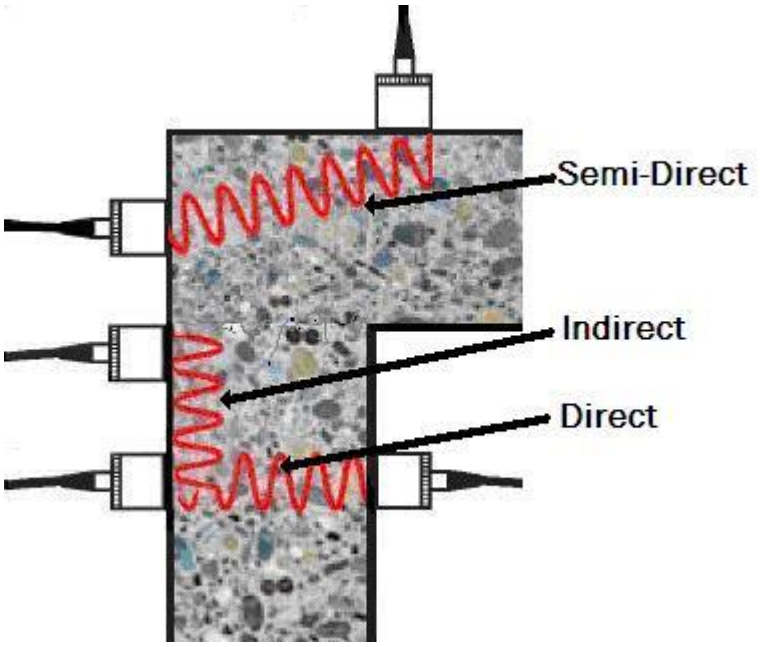


Figure 1.6 Transducer Arrangements

2. Determination of pulse velocity by direct transmission

Where possible the direct transmission arrangement should be used since the transfer of energy between transducers is at its maximum and the accuracy of velocity determination is therefore governed principally by the accuracy of the path length measurement. The couplant used should be spread as thinly as possible to avoid any end effects resulting from the different velocities in couplant and concrete.

3. Determination of pulse velocity by semi-direct transmission

The semi-direct transmission arrangement has a sensitivity intermediate between those of the other two arrangements and, although there may be some reduction in the accuracy of measurement of the path length, it is generally found to be sufficiently accurate to take this as the distance measured from centre to centre of the transducer faces. This arrangement is otherwise similar to direct transmission.

4. Determination of pulse velocity by indirect or surface transmission

Indirect transmission should be used when only one face of the concrete is accessible, when the depth of a surface crack is to be determined or

when the quality of the surface concrete relative to the overall quality is of interest. It is the least sensitive of the arrangements and, for a given path length, produces at the receiving transducer a signal which has an amplitude of only about 2% or 3% of that produced by direct transmission. Furthermore, this arrangement gives pulse velocity measurements which are usually influenced by the concrete near the surface. This region is often of different composition from that of the concrete within the body of a unit and the test results may be unrepresentative of that concrete. The indirect velocity is invariably lower than the direct velocity on the same concrete element. This difference may vary from 5% to 20% depending largely on the quality of the concrete under test. Where practicable site measurements should be made to determine this difference. With indirect transmission there is some uncertainty regarding the exact length of the transmission path because of the significant size of the areas of contact between the transducers and the concrete. It is therefore preferable to make a series of measurements with the transducers at different distances apart to eliminate this uncertainty. To do this, the transmitting transducer should be placed in contact with the concrete surface at a fixed point x and the receiving transducer should be placed at fixed increments x_o along a chosen line on the surface. The transmission times recorded should be plotted as points on a graph showing their relation to the distance separating the transducers. The slope of the best straight line drawn through the points should be measured and recorded as the mean pulse velocity along the chosen line on the concrete surface. Where the points measured and recorded in this way indicate a discontinuity, it is likely that a surface crack or surface layer of inferior quality is present and a velocity measured in such an instance is unreliable.

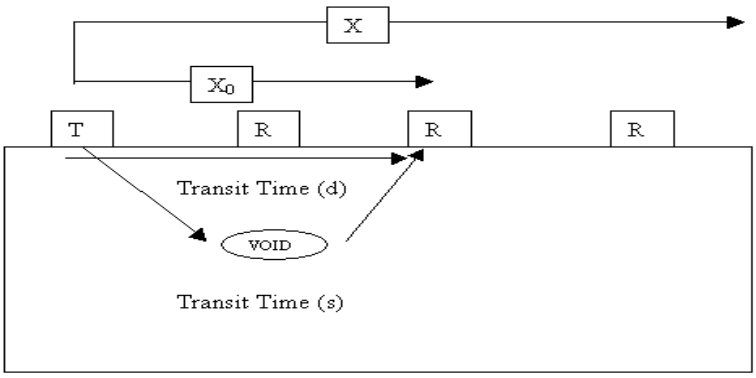


Figure 1.7 Pulse Velocity Determinations by Indirect (Surface) Transmission

5. Coupling the transducer onto the concrete

To ensure that the ultrasonic pulses generated at the transmitting transducers pass into the concrete and are then detected by the receiving transducer, it is essential that there is adequate acoustical coupling between the concrete and the face of each transducer. For many concrete surfaces, the finish is sufficiently smooth to ensure good acoustical contact by the use of a coupling medium and by pressing the transducer against the concrete surface. Typical couplants are petroleum jelly, grease, soft soap and kaolin/glycerol paste. It is important that only a very thin layer of coupling medium separates the surface of the concrete from its contacting transducer. For this reason, repeated readings of the transit time should be made until a minimum value is obtained so as to allow the layer of the couplant to become thinly spread. Where possible, the transducers should be in contact with the concrete surfaces, which have been cast against formwork or a mold. Surfaces formed by other means, e.g. trowelling, may have properties differing from those of the main body of material. If it is necessary to work on such a surface, measurements should be made over a longer path length than would normally be used. A minimum path length of 150 mm is recommended for direct transmission involving one unmolded surface and a minimum of 400 mm for indirect transmission along one unmolded surface. When the concrete surface is very rough and uneven, the area of the surface where the transducer is to be applied should be smoothed and leveled. Alternately, a smoothing medium such as quick setting epoxy resin or plaster may be used, but good adhesion between the concrete surface and the smoothing medium has to be ensured so that the pulse propagates correctly into the concrete under test. It is important to ensure that the layer of smoothing medium is as thin as possible. If it is necessary to make a significant build up then the pulse velocity of the smoothing medium has to be taken into account.

- **Factors Influencing Pulse Velocity Measurements**

1. Moisture content

The moisture content has two effects on the pulse velocity, one chemical the other physical. These effects are important in the production of

correlations for the estimation of concrete strength. Between a properly cured standard cube and a structural element made from the same concrete, there may be a significant pulse velocity difference. Much of the difference is accounted for by the effect of different curing conditions on the hydration of the cement while some of the difference is due to the presence of free water in the voids. It is important that these effects are carefully considered when estimating strength.

2. Path length

The path length over which the pulse velocity is measured should be long enough not to be significantly influenced by the heterogeneous nature of the concrete. It is recommended that, the minimum path length should be 100 mm for concrete where nominal maximum size of aggregate is 20 mm or less and 150 mm for concrete where nominal maximum size of aggregate is between 20 mm and 40 mm. The pulse velocity is not generally influenced by changes in path length, although the electronic timing apparatus may indicate a tendency for velocity to reduce slightly with increasing path length. This is because the higher frequency components of the pulse are attenuated more than the lower frequency components and the shape of the onset of the pulse becomes more rounded with increased distance traveled. Thus, the apparent reduction of pulse velocity arises from the difficulty of defining exactly the onset of the pulse and this depends on the particular method used for its definition. This apparent reduction in velocity is usually small and well within the tolerance of time measurement accuracy for the equipment.

3. Shape and size of specimen

The velocity of short pulses of vibration is independent of the size and shape of the specimen in which they travel, unless its least lateral dimension is less than a certain minimum value. Below this value, the pulse velocity may be reduced appreciably. The extent of this reduction depends mainly on the ratio of the wavelength of the pulse vibrations to the least lateral dimension of the specimen but it is insignificant if the ratio is less than unity. Table 1.2 gives the relationship between the pulse velocity in the concrete, the transducer frequency and the minimum permissible lateral dimension of the specimen. If

the minimum lateral dimension is less than the wavelength or if the indirect transmission arrangement is used, the mode of propagation changes and therefore the measured velocity will be different. This is particularly important in cases where concrete elements of significantly different sizes are being compared.

TABLE 1.1 Effectsof specimen dimensions on pulse transmission

Transducer frequency	**Pulse velocity in concrete (km/s)**		
KHz	V_c	V_c	V_c
	Minimum permissible lateral specimen dimension		
	mm	mm	mm
24	146	167	188
54	65	74	83
82	43	49	55
150	23	27	30

4. Effect of reinforcing bars

The pulse velocity measured in reinforced concrete in the vicinity of reinforcing bars is usually higher than in plain concrete of the same composition. This is because the pulse velocity in steel may be up to twice the velocity in plain concrete and, under certain conditions, the first pulse to arrive at the receiving transducer travels partly in concrete and partly in steel. The apparent increase in pulse velocity depends on the proximity of the measurements to the reinforcing bar, the diameter and number of bars and their orientation with respect to the propagation path. The frequency of the pulse and surface conditions of the bar may both also affect the degree to which the steel influences the velocity measurements. Corrections to measured values to allow for reinforcement will reduce the accuracy of estimated pulse velocity in the concrete so that, wherever possible, measurements should be made in such a way that steel does not lie in or close to the direct path between the transducers.

5. Temperature of the concrete

Variations of the concrete temperature between 10ºC and 30ºC have been found to cause no significant change without the occurrence of corresponding changes in the strength or elastic properties. Corrections to pulse velocity measurements should be made only for temperatures outside this range as given in Table 1.2.

TABLE 1.2 Effect of temperature on pulse transmission

Temperature	Correlation to the measured pulse velocity	
	Air dried concrete	Water saturated concrete
ºC	%	%
60	+5	+4
40	+2	+1.7
20	0	0
0	-0.5	-1
-4	-1.5	-7.5

- **Determination of Concrete Uniformity**

Heterogeneities in the concrete within or between members cause variations in pulse velocity, which in turn are related to variations in quality. Measurements of pulse velocity provide a means of studying the homogeneity and for this purpose a system of measuring points which covers uniformly the appropriate volume of concrete in the structure has to be chosen.The number of individual test points depends upon the size of the structure, accuracy required and variability of the concrete. In a large unit of fairly uniform concrete, testing on a 1m grid is usually adequate but, on small units or variable concrete, a finer grid may be necessary. It should be noted that, in cases where the path length is the same throughout the survey, the measured time might be used to assess the concrete uniformity without the need to convert it to velocity. This technique is particularly suitable for surveys where all the measurements are made by indirect measurements. It is possible to express homogeneity in the form of a statistical parameter such as the standard deviation or coefficient of variation of the pulse velocity measurements made over a grid. However, such parameters can only be properly used to compare

variations in concrete units of broadly similar dimensions. Variations in pulse velocity are influenced by the magnitude of the path length because this determines the effective size of the concrete sample, which is under examination during each measurement. The importance of variations should be judged in relation to the effect which they can be expected to have on the required performance of the structural member being tested. This generally means that the tolerance allowed for quality distribution within members should be related either to the stress distribution within them under critical working load conditions or to exposure conditions.

- **Detection of Defects**

The use of the ultrasonic pulse velocity technique to detect and define the extent of internal defects should be restricted to well-qualified personnel with previous experience in the interpretation of survey results. Attention is drawn to the potential risk of drawing conclusions from single results. When an ultrasonic pulse travelling through concrete meets a concrete-air interface there is negligible transmission of energy across this interface. Thus any air filled void lying immediately between transducers will obstruct the direct ultrasonic beam when the projected length of the void is greater than the width of the transducers and the wavelength of sound used. When this happens the first pulse to arrive at the receiving transducer will have been diffracted around the periphery of the void and the transit time will be longer than in similar concrete with no void. It is possible to make use of this effect for locating flaws, voids or other defects greater than about 100 mm in diameter or depth. Relatively small defects have little or no effect on transmission times but equally are probably of minor engineering importance. Plotting contours of equal velocity often gives significant information regarding the quality of a concrete unit. The method used to detect a void is to draw a grid on the concrete with its points of intersection spaced to correspond to the size of void that would significantly affect the concrete performance. A survey of measurements at the grid points enables a large cavity to be investigated by measuring the transit times of pulses passing between the transducers when they are placed so that the cavity lies in the direct path between them. The size of such cavities may be estimated by assuming that the pulses pass along the

shortest path between the transducers and around the cavity. Such estimates are valid only when the concrete around the cavity is uniformly dense and the pulse velocity can be measured in that concrete. The method is not very successful when applied to structures with cracks because the cracked faces are usually sufficiently in contact with each other to allow the pulse energy to pass unimpeded across the crack. This can happen in cracked vertical bearing piles where there is also sufficient compression to hold the faces close together. If the concrete is surrounded by water such that the crack is filled with water, the crack is undetectable since ultrasonic energy can travel through a liquid.

1.5 History Development In Testing

In recent years, a lot of studies on built heritage emphasize the need to use appropriate techniques to evaluate the current condition of the structure before designing an intervention. In fact, the technicians very often encounter a lot of economic and logistical problems that are barriers to planning o suitable diagnostic tests; and that need consideration with regards to the masonry buildings or reinforced concrete buildings. The research focus on the restoration of reinforced concrete built buildings, that being to show signs of decay and deterioration.

In reinforced concrete structure, the most important properties are-among others-the concrete compressive strength; the concrete condition and steel bars yield strength and fracture. Through the concrete compressive strength, other concrete properties can be obtained, such as elastic modulus, tensile strength and durability. So to do this, the “destructive” methods an “non-destructive” test such as the sclerometer test, the ultrasonic test and the combined method called “SonReb” can be used.

The combined results from different “non-destructive” test are very interesting to assess the concrete strength. This methodological approach can help to reduce the possible errors when using the sclerometer and ultrasonic tests separately; in this combined method called “SonReb” (SONic + REBound) were developed. In fact, it has been noticed that humidity content of a structural element can influence the scelerometer index and at the same time the ultrasonic speed. Using a combined method it is possible to balance the sum of errors made when two methodologies used

separately. Moreover, the combined method requires short time to obtain the result, it's a non invasive method and it does not affect the resistance of structural elements.

The above defined methodology has been tested on a post-war building which is located in Puglia (Italy): the "Centron" Theatre; it was built a mixed structure i.e. reinforced concrete and bearing masonry built of local stone. The building used until 90's and now is abandoned

CHAPTER - 2

LITERATURE REVIEW

The comprehensive review of literature shows that a considerable amount of work related to the determination of compressive strength of concrete specimen by Non-destructive testing is done worldwide.

Samiahannachi, Mohamed Nacerguettech[1]:

It appears that more than one non-destructive technique provides a better correlation and the sense contributes to more reliable strength evaluation of concrete. The combined method seems more promising to evaluate the compressive strength of concrete in construction. It will also be noted that correlation between destructive testing an non-destructive techniques in our study provide more meaningful results for the specimen cast and stored under the same condition as the concrete structure than takes cores.

The practical use of this technique is gaining recognition on a large scale, it provides contracting authorize with accurate and objective information for monitoring quality-control of concrete construction. And the combined method appears more appropriate to condition on sitemeasurements, veryconvenient, fast and with a reasonable cost.

Suresh Chandra Pattanaik[2]:

The structural health monitoring by NDT methods such as UPV and Rebound Hammer becomes very useful for predicting the service life of structure and deterioration of the structure provides the periodical monitoring of the same member of the structure is being carried out. since the concrete is heterogeneous and test are affected by various factors such as age of concrete, carbonation depth, reinforcement, cracks and voids inside the concrete, a combined test help for assessment the strength and durability. The experimental investigation showed that a good co-relation exists between compressive strength, rebound hammer and ultrasonic pulse velocity. Nevertheless rebound hammer should be used alone to determine the compressive strength of structures. Ultrasonic Pulse Velocity is the ideal NDT method.

M. T. Cristofaro[3]:

The evaluation of the concrete compressive strength is a fundamental step for the assessment of existing reinforcement concrete according to the most recent seismic codes. Non-Destructive method, although being minimally invasive and easily to a large number of element, are influenced by many factors.

The comparison carried out shows destructive methodologies associated with a non-destructive method allows obtaining higher level of knowledge and a greater accuracy on the estimation of the concrete compressive strength if the relationship is calibrated on the individual building. The results using the SonReb method, calibrated with the strength of cylindrical specimen extracted from a single building are close to actual ones.

J. C. Agunwamba[4]:

The results for hardened concrete show a reasonable correlation of correlation of compressive strength with the rebound hammer and ultrasonic pulse velocity. The sensitivity of the pulse velocity test in measuring strength is affected by the concrete age, as the concrete mature, the sensitivity of Ultrasonic Pulse Velocity to strength gain or achieved by the concrete increases. The rebound hammer shows less sensitivity as the concrete mature since it is surface hardness test and for age above 7 days there is little on no gain in surface hardness.

Mohammadreza Mahmoudipour[5]:

Nondestructive tests are used to evaluate the serviceability and safety of the structures. Schimidt Hammer and Ultrasonic Pulse Velocity tests are very common in evaluating concrete strength. These NDT's are not only giving a criteria for retrofitting plan, but also are not affect the integrity of the structures according to different concrete parameters such as aggregate, cement, different placing method, etc.

In order to check serviceability and strength of structure, non-destructive tests are used to increase the speed of evaluation. Moreover,NDT would not affect the integrity of the structure and very large no of spot reading can be done with very low cost in comparison with destructive tests. Schmidth hammer tests and ultrasonic tester are very convenient and can execute anywhere.

Dallshad K.H. Bzeni[6]:

In this paper nondestructive test methods such as rebound hammer, ultrasonic pulse velocity and combined method are correlated with destructive test methods, crushing strength of SCC cubes. Six mixes of SCC were prepared, with different starting from cementations material content 297 up to 500 kg/m^3. Flow ability and passing ability of fresh SCC mixes were approved.

On the basis of the experimental results obtained in this work, using two diff NDT methods for predicting compressive strength of SCC, the following conclusion can be withdrawn

1. Schmidt hammer reading are less sensitive to assessed compressive strength of SCC compared to conventional vibrated concrete.
2. As a general trend increasing in water to cementitious material ratio decreases rebound number and UPV values, while there is no generalized formula that can be used because of low correlation coefficients resulted.
3. Introducing information about the age of concrete at the time of test with rebound number of hammer test into the regression equation, include the predicted value of compressive strength of SCC.

Roberto C. A. Pinto[7]:

This study discusses the use of ultrasound with the time of flight diffraction technique to estimate the depth of surface opening cracks,. This technique is very useful tool for practical applications. It is expensive simple and easy to perform, giving a rapid indication of the extension of cracking. BS 1881 part 203 presents some mathematical expressions to estimate the depth of surface opening cracks.

Two graphically based methods have been developed to estimate cracks penetration depth from time of flight measurement. The results indicated that the method developed as an extension of the one in the BS 1881 standard with at least four readings was able to improve the overall estimation with an estimated error close to 10%, smaller than the errors from the other methods. This method also permits the estimation of the ultrasound pulse velocity in the region tested.

CHAPTER - 3

METHODOLOGY AND INVESTIGATIONS

3.1 Ingredients and Testing Of Ingredients

3.1.1 Ingredients

- Cement
- Water
- Fine aggregate
- Coarse aggregate

Percentage of ingredients in a mix:

Cement - 7% to 17%

Water - 15% to 20%

Aggregate (coarse and fine) - 78% to 63%

Paste (water + cement) - 22% to 37%.

It should be noted that the aggregates in the concrete mix constitute by far the bulk of the mass. The properties of the concrete produced depend upon the amount and type of materials used, and the way they are mixed, handled, compacted, finished and cured.

3.1.2 Properties

Cement - There are many types of cement available for making concrete. Some countries have a national standard, which prescribes the requirement for any cement manufactured or used in that country. e.g. Australian Standard AS3972-1991 “Portland and Blended Cements”. Each type of cement produced has a different chemical composition and fineness and gives different properties to a concrete.

There are four main types:

1. General purpose cement - the most common and the one used for the majority of structures,
2. Type HE - high early strengths cement - gains strength quickly,

3. Type LH - low Heat Cement - produces less heat than GP and Type HE cement. It gains strength more slowly,

4. Type SR - sulphate resisting cement - cement that resists sulphate attack.

Aggregates - Coarse aggregates are stones that are more than 5 mm in diameter and are either crushed rock generally from quarries, or gravels excavated from pits or dredged from river beds. Fine aggregates consist of fine and coarse sands or crushed rock finer than 5 mm. The aggregates used should cover a range of sizes. They should be clean and free of any contaminating substances, which may adversely affect the setting time, strength or durability of the concrete, or corrode the steel reinforcement.

Aggregates should not contain:

- Weak substances such as pieces of wood, humus or coal
- Dirt, clay dust or silt coatings.

These reduce the bond between aggregates and the cement paste

- Water soluble salts such as sulphates or chlorides.

Aggregates should be strong, hard and durable in order to develop the full strength of the cement paste and maximum wear resistance of the concrete. Crumbly or flaky rocks like sandstone, slates or shales should not be used since they lack strength. Chert aggregates should not be used to avoid alkali-silica reaction.

The shape of the aggregate particles is also important since the shape affects both the workability and strength of the concrete. Smooth rounded aggregates produce workable and easy to handle concrete. Angular materials tend to give a stronger concrete but reduce workability. Flaky and elongated materials promote segregation and reduce workability. They also require more sand and cement. The use of these aggregates should be limited.

There are certain types of aggregates that can react with cement in the presence of moisture. Such reactions result in expansive compounds, which crack and deteriorate the concrete. Suspect aggregates should not be used.

Mixing water - Water, which is necessary for hydration, must be clean and fresh and not contain any impurities since these may affect the concrete properties. It is generally accepted that water, which is fit for drinking, is suitable for making concrete. Seawater should not be used in making concrete, particularly reinforced concrete, as it will corrode the steel reinforcement. Bore water must be analyzed first. Industrial, waste or brackish water should not be used. Sugars and detergents have a retarding effect on the setting properties of concrete and, therefore, should not be added to a concrete mix.

1. **Cement**:-

   Ordinary Portland cement, 43 grade can be used.

2. **Water**:-

   Water should be pure. It should be usable for drinking purpose.

3. **Fine Aggregate**:-

   The grading must be uniform throughout the work. The moisture content or absorption characteristics must be closely monitored as quality of LWC will be sensitive to such changes. Size of fine aggregate should be bigger than 125 micron and smaller than 4.75mm.

4. **Coarse Aggregate**:-

   The maximum size of aggregate is generally limited to 20mm. Aggregate of maximum size 10-12mm is desirable for structures having congested reinforcement.

3.2 EXPERIMENTAL WORK

The experimental work includes the casting, curing and testing of concrete. A concrete mix of M15 to M45 grade is designed. The locally available fine aggregate and coarse aggregate are used.

3.2.1 TESTING OF MATERIAL USED FOR DIFFERENT GRADES OF CONCRETE

The material used for project work such as cement, coarse aggregate and fine aggregate is initially checked which is explained as below,

3.2.1.1 CEMENT

Ordinary Portland cement of 43 grade is use in this project work. This specific gravity is 3.15.

1. Fineness Test:

Weight of cement = 100gm

Weight retained on IS 90μ sieve after 15min. hand shaking = 7 gm.

Percentage Residue by weight of the sample = 7%

As per IS: 4031-1968, the residue should not exceed 7% by weight of the cement sample. The test result is below the limiting value.

2. Standard Consistency Test:

The Standard Consistency of the cement is 28.5%.

3. Setting Time Test:

Initial setting time of cement = 130 min. (> 30 min.)

Final setting time of cement = 205 min. (< 600 min.)

4. Compressive Strength Test of Cement Mortar Cubes:

TABLE 3.1 Cement Mortar Cube Compressive Strength

Sr. No.	Curing Period	Average Comp. Strength (N/mm^2)	Values According to IS: 12269-2009
1.	28 Days	52.5	> 43

Results of compressive strength of mortar cubes are as per IS specifications. So this cement is OK to use in concrete for experimental study.

5. Soundness Test:

Difference between the two pointers should not exceed 10 mm as per Indian Standard. The soundness of the cement sample is found to be 1 mm. The test results falls within the limit.

3.2.1.2 COARSE AGGREGATE

Various tests on coarse aggregate are done as per IS 383:1970.

1. Specific Gravity of coarse aggregate (10mm-20mm)

Weight of aggregate with bucket immersed in water, W_1 =1.3 kg

Weight of bucket immersed in water, W_2= 0.841 kg

Weight of saturated surface dried aggregate, W_3 = 2.410kg

(After this aggregate place in oven dried for 24 Hrs)

Oven dried Weight of aggregate, W_4= 1.516kg

$$\text{Sp. gravity} = \frac{W4}{W3 - (W1 - W2)}$$

$$\text{Sp. gravity} = \frac{1.516}{2.410 - (1.3 - 0.841)}$$

$$\text{Sp. gravity} = 0.71$$

2. Specific Gravity of coarse aggregate (4.75mm-10mm)

$$\text{Sp. gravity} = 0.65$$

3. Water absorption:-

$$\text{Water absorption} = \frac{W3 - W4}{W4} \text{X } 100$$

$$\text{Water absorption} = \frac{2.410 - 1.516}{1.516} \text{X } 100$$

$$\text{Water absorption} = 58.9\%$$

4. Sieve Analysis:

TABLE 3.2 Sieve Analyses for Coarse Aggregate

IS Sieve	Weight Retained (kg)	Cumulative Weight Retained	Cumulative Percentage Retained (%)
63 mm	0	0	0
40 mm	0	0	0
20 mm	0.524	0.524	26.2
10 mm	1.434	1.958	97.9
4.75 mm	0.034	1.992	99.6
2.0mm	0.008	2.0	100
1.18 mm	0	0	100
600 µ	0	0	100
300 µ	0	0	100
150µ	0	0	100
Pan	0	0	100

$$\text{Fineness modulus} = \frac{\text{Total Comulative Percentage Retained}}{100}$$

$$\text{Finess modulus} = \frac{323.7}{100}$$

Fineness modulus of coarse aggregate = 3.237%



Figure 3.1 Sieve Analysis for Coarse Aggregate

3.2.1.3 FINE AGGREGATE

Various tests on fine aggregate are done as per IS 383:1970. Natural river sand is use.

1. Specific Gravity (by Pycnometer):

Weight of empty Pycnometer = W_1= 0.640kg

Weight of empty Pycnometer and sand = W_2 =1.070 kg

Weight of empty Pycnometer, sand and water = W_3 =1.680 kg

Weight of empty Pycnometer and water = W_4 = 1.450kg

$$\text{Specific Gravity} = \frac{W_2 - W_1}{(W_4 - W_1) - (W_3 - W_2)}$$

$$\text{Specific Gravity} = \frac{1.070 - 0.640}{(1.450 - 0.640) - (1.680 - 1.070)}$$

$$\text{Specific Gravity of Fine Aggregate} = 2.15$$

2. Sieve Analysis:

TABLE 3.3 Sieve Analyses for Fine Aggregate

IS Sieve	Weight Retained (kg)	Cumulative Weight Retained (kg)	Cumulative Percentage Weight Retained (%)
80 mm	0	0	0
40 mm	0	0	0
20 mm	0	0	0
10 mm	0	0	0
4.75 mm	0.28	0.28	14
2.0 mm	0.53	0.81	40.5
1.0mm	0.48	1.29	64.5
600 µ	0.28	1.57	78.5

300 µ	0.37	1.94	97.0
Pan	0.06	2	100

$$\text{Fineness modulus} = \frac{\text{Total Cummulative Percentage Retained}}{100}$$

$$\text{Fineness modulus} = \frac{280.5}{100}$$

$$\text{Fineness modulus of fine aggregate} = 2.805\%$$

As per IS 383:1970, Grading zone is I.

3. Water absorption (by Oven Drying Method):

Weight of saturated surface dried sand, W_1= 1.15 kg

Weight of oven dried sand, W_2 = 0.960 kg

$$\text{Water Absorption} = \frac{(W_1 - W_2)}{W_2} \times 100$$

$$\text{Water Absorption} = \frac{(1.15 - 0.960)}{0.960} \times 100$$

$$\text{Water Absorption of fine aggregate} = 1\%$$

3.2.1.4 Water:

Clean potable water was used for mixing.

3.3 Preparation of Test Specimens:-

For casting of the cube specimens, moulds of size (150 x 150x 150) mm are used. A total of 60 concrete specimen was designed and fabricated

Step.1.Batching:-

Weight batching is used for preparing the concrete mix. The grade of concrete is used M15, M20,M25,M30,M35,and M40. The ratio of cement, sand and coarse aggregate is obtained by the mix design done above.In particular 10 specimens of each grade.

Step.2.Mixing:-

Hand mixing over a mixing tray is done throughout by taking required quantities of cement, sand, coarse aggregate. Coarse Aggregate were placed first in the tray followed by crush sand and then cement. All these ingredients are first mixed in dry condition for 1min, then calculated amount of water is added and mixing is continued until uniform mix obtained.



Figure 3.2 Mixing of Concrete

Step.3.Placing:-

Before placing concrete in the mould, it is kept ready by applying oil inside it. After proper mixing of concrete, material is filled in mould, in three layers. Each layer is compacted by standard tamping rod with more than 35 strokes. Exposed surface was finished with trowel to avoid uneven surface



Figure 3.3 Oiling the moulds before placing the concrete

Step.4.Curing:-

The concrete specimens are removed from the mould after 24 hrs. All the concrete specimens are kept in water for 28 days curing.



Figure 3.4 Curing Tank

CHAPTER - 4

RESULT AND DISCUSSIONS

4.1 Mix Proportion and Concrete Batches

TABLE 4.1 Proportion and Batches

Grade	Cement (kg/m^3)	Aggregate (kg/m^3)		Water (lit/m^3)
		Fine	Coarse	
M15	270	711	1460	135
M20	320	794	1138	176
M25	340	775	1113	185
M30	380	760	1090	187
M35	410	735	1053	200
M40	430	718	1030	185

4.2 Interpretation of Results

The results obtained from all the CTM, Rebound Hammer and UPV, are thus interpreted in the following manner. Among the obtained 60 results, 20 random results are selected.

TABLE 4.2 Observed Readings of 60 Specimens

R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}	R_{avg}	UPV	Fck
24	**22**	**22**	**24**	**22**	**22**	**24**	**22**	**24**	**24**	**23**	**3620**	**16.8**
20	24	24	22	22	24	20	22	20	26	22.4	4259	16.9
24	22	24	24	22	24	26	24	24	24	23.8	3960	17.2
24	**26**	**24**	**24**	**28**	**22**	**24**	**24**	**28**	**24**	**24.8**	**3718**	**17.4**
24	22	24	22	24	26	22	22	24	24	23.4	3612	17.8
22	24	22	24	26	24	26	26	22	24	24	4130	17.9
24	**22**	**26**	**24**	**24**	**26**	**24**	**28**	**30**	**24**	**25.2**	**3906**	**18.1**
28	24	28	26	24	26	26	24	24	26	25.6	4003	18.28
26	24	26	28	26	24	24	26	28	26	25.8	4159	18.6
26	**26**	**24**	**26**	**30**	**24**	**28**	**28**	**26**	**30**	**26.8**	**3789**	**19.6**
30	30	26	24	24	28	26	28	26	26	26.8	3469	19.7
26	28	26	26	30	30	26	26	26	28	27.2	4259	20
26	**24**	**28**	**28**	**26**	**30**	**28**	**28**	**32**	**26**	**27.6**	**4003**	**20.1**
24	26	26	26	24	24	28	26	26	26	25.6	3911	20.1
28	28	26	28	28	30	26	28	30	26	27.8	3150	20.3
26	**28**	**26**	**26**	**28**	**28**	**26**	**30**	**28**	**28**	**27.4**	**4112**	**20.4**
28	30	26	28	30	28	30	28	28	28	28.4	3462	20.5
32	32	28	26	28	32	29	26	30	27	29	3752	20.9
28	**30**	**28**	**32**	**32**	**28**	**30**	**28**	**32**	**28**	**29.6**	**3964**	**21.4**
30	30	32	30	30	28	32	30	28	30	30	4003	22.3
32	28	30	30	28	32	30	32	30	30	30.2	3625	23.5
28	**26**	**34**	**30**	**32**	**32**	**34**	**34**	**34**	**34**	**31.8**	**4003**	**23.8**
30	30	34	32	36	30	32	28	32	34	31.8	4259	24.2
28	32	30	34	34	34	32	36	32	32	32.4	3960	24.4
32	**32**	**32**	**36**	**34**	**32**	**32**	**34**	**34**	**32**	**33**	**3998**	**25.6**
34	30	36	34	32	34	34	30	34	32	33	3118	26.3
30	34	30	34	34	34	32	36	30	32	32.6	3620	26.5
32	**36**	**34**	**32**	**32**	**32**	**34**	**30**	**32**	**28**	**32.2**	**4112**	**26.9**
28	30	28	30	28	28	30	28	28	28	28.6	3589	27.4

30	30	26	28	30	28	26	30	28	30	28.6	4259	27.8
30	**32**	**30**	**28**	**28**	**30**	**30**	**28**	**30**	**26**	**29.2**	**4049**	**28.9**
26	28	28	28	26	30	26	26	28	30	27.6	3742	29.17
28	30	28	32	32	28	30	28	32	30	29.8	3960	29.2
28	**32**	**32**	**30**	**28**	**28**	**32**	**28**	**30**	**30**	**29.8**	**4109**	**29.6**
26	28	28	30	28	30	26	26	30	30	28.2	4159	29.8
28	26	26	28	30	28	30	28	28	28	28	4003	29.9
30	**28**	**30**	**32**	**28**	**32**	**32**	**30**	**30**	**32**	**30.4**	**4112**	**30.3**
28	28	34	28	32	34	34	34	34	32	31.8	3625	30.7
30	32	30	32	28	28	32	30	30	30	30.2	3841	31.1
28	**32**	**30**	**34**	**34**	**36**	**32**	**36**	**30**	**32**	**32.4**	**4129**	**31.2**
36	34	30	34	34	30	34	30	32	34	32.8	3625	31.6
30	28	32	28	32	30	30	28	30	32	30	4257	32.3
28	**32**	**28**	**32**	**30**	**32**	**34**	**32**	**32**	**32**	**31.2**	**4219**	**32.4**
34	34	30	32	32	30	34	34	34	30	32.4	4259	32.56
30	30	30	32	30	30	34	32	30	32	31	4085	32.8
28	**32**	**34**	**32**	**34**	**30**	**34**	**28**	**32**	**34**	**31.8**	**4199**	**33.1**
32	32	32	34	34	34	32	36	32	28	32.6	3342	33.45
34	30	32	34	32	30	34	30	32	34	32.2	3210	34.4
34	**30**	**36**	**34**	**36**	**32**	**32**	**34**	**34**	**34**	**33.6**	**3394**	**36.4**
34	32	34	34	28	34	34	30	34	34	32.8	4398	36.1
32	30	36	34	36	32	30	34	34	34	33.2	3394	36.4
34	**34**	**32**	**36**	**34**	**34**	**34**	**36**	**36**	**32**	**34.2**	**4159**	**36.9**
32	34	32	36	36	32	36	36	36	36	34.6	4187	37.2
32	32	30	32	34	32	36	36	30	30	32.4	3960	37.8
36	34	32	30	32	34	34	32	36	34	33.4	4110	38.2
34	**36**	**34**	**36**	**36**	**38**	**36**	**36**	**36**	**36**	**35.8**	**4259**	**38.4**
36	36	36	36	38	42	36	36	36	38	37	3967	38.6
38	42	36	36	36	36	36	42	36	40	37.8	3625	39.3
40	34	36	40	40	38	36	40	40	42	38.6	4112	39.8
42	**40**	**40**	**42**	**38**	**40**	**40**	**40**	**36**	**40**	**39.8**	**4159**	**40.2**

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TEST REPORT

TR-1504070072

Non- Destructive test of concrete structure

Sample ID	:	NDT-136	Report No	:	NDT-136/1
Name of Customer	:	Group No. 2, BE CIVIL	Date of receipt	:	07/04/2015
Office Address	:	G S Moze College of Engineering, Pune.	Date of testing	:	03/03/2015
			Date of Report	:	07/04/2015
Site Name	:		Sample condition	:	Acceptable
Sample Description	:	Concrete Cubes	Witnessed By	:	All Group members
Grade of Concrete	:	-	Age of Concrete	:	More than 28 Days

NDT test by UPV and Rebound Hammer method as per IS 13311 (P-I & II) : 1992 Reaffirmed 2008

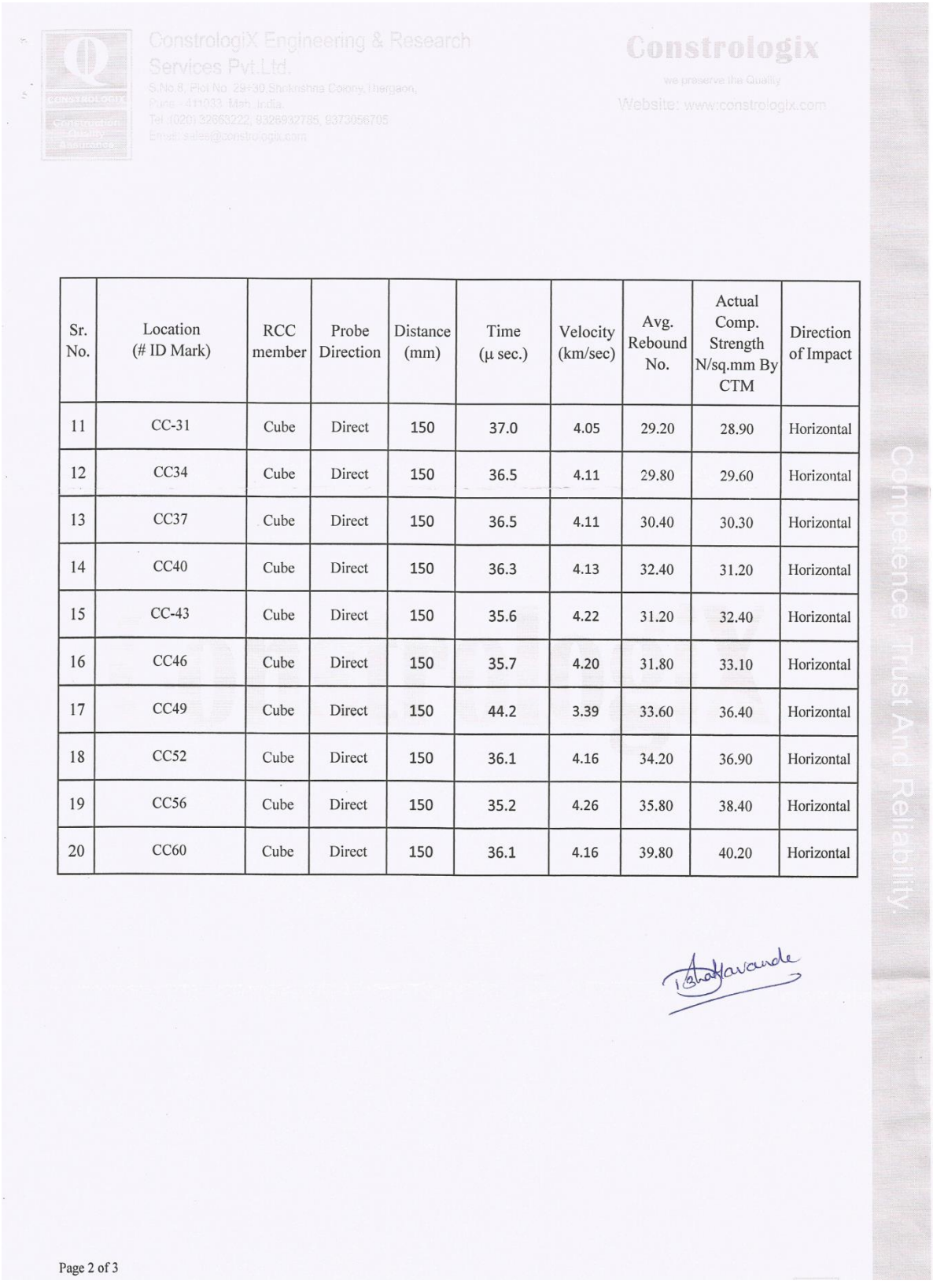
Sr. No.	Location (# ID Mark)	RCC member	Probe Direction	Distance (mm)	Time (μ sec.)	Velocity (km/sec)	Avg. Rebound No.	Actual Comp. Strength N/sq.mm By CTM	Direction of Impact
1	CC-01	Cube	Direct	150	41.4	3.62	23.00	16.80	Horizontal
2	CC-04	Cube	Direct	150	40.3	3.72	24.80	17.40	Horizontal
3	CC-07	Cube	Direct	150	38.4	3.91	25.20	18.10	Horizontal
4	CC-10	Cube	Direct	150	39.6	3.79	26.80	19.60	Horizontal
5	CC-13	Cube	Direct	150	37.5	4.00	27.60	20.10	Horizontal
6	CC-16	Cube	Direct	150	36.5	4.11	27.40	20.40	Horizontal
7	CC-19	Cube	Direct	150	37.8	3.96	29.60	21.40	Horizontal
8	CC-22	Cube	Direct	150	37.5	4.00	31.80	23.80	Horizontal
9	CC-25	Cube	Direct	150	37.5	4.00	33.00	25.60	Horizontal
10	CC-28	Cube	Direct	150	36.5	4.11	32.20	26.90	Horizontal

Refer Drawing _-___for tests locations.

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Competence, Trust And Reliability.

Figure 4.1(a) Results by Laboratory



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Sr. No.	Location (# ID Mark)	RCC member	Probe Direction	Distance (mm)	Time (μ sec.)	Velocity (km/sec)	Avg. Rebound No.	Actual Comp. Strength N/sq.mm By CTM	Direction of Impact
11	CC-31	Cube	Direct	150	37.0	4.05	29.20	28.90	Horizontal
12	CC34	Cube	Direct	150	36.5	4.11	29.80	29.60	Horizontal
13	CC37	Cube	Direct	150	36.5	4.11	30.40	30.30	Horizontal
14	CC40	Cube	Direct	150	36.3	4.13	32.40	31.20	Horizontal
15	CC-43	Cube	Direct	150	35.6	4.22	31.20	32.40	Horizontal
16	CC46	Cube	Direct	150	35.7	4.20	31.80	33.10	Horizontal
17	CC49	Cube	Direct	150	44.2	3.39	33.60	36.40	Horizontal
18	CC52	Cube	Direct	150	36.1	4.16	34.20	36.90	Horizontal
19	CC56	Cube	Direct	150	35.2	4.26	35.80	38.40	Horizontal
20	CC60	Cube	Direct	150	36.1	4.16	39.80	40.20	Horizontal

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Competence, Trust And Reliability.

Figure 4.1(b) Reports by Laboratory

4.2.1 Compression Testing Machine

The accuracy of the testing machine shall be such that the percentage of error for the loads within the proposed range of use of the testing machine shall not exceed ± 1.0 percent of the indicated load. The indicated load of a testing machine shall not be corrected either by calculation or by the use of calibration diagram to obtain values within the required permissible variation.

The measured compressive strength of the specimen shall be calculated by dividing the maximum load applied to the specimen during the test by the cross-sectional area. calculated for all the mean dimensions of the section *(* IS· 1199-1959) and shall be expressed to the nearest kg per sq. cm, Average of three values shall be taken as the representative of the batch provided the Individual variation is not more than ± 15 percent of the average. Otherwise repeat tests shall be made.

Compressive testing machine of capacity 2000 KN is used for compression testing of cube as casted of size 150 x 150 x 150 mm and capable of giving load at the rate of 140 kg/sq.cm/min. The wet cubes were placed in the machine between wiped and cleaned loading surface and load is given approximately at the rate of 140 kg/sq.cm/min.



Figure 4.2 Compression Testing Machine

4.2.2 Rebound Hammer.

The rebound hammer method provides a convenient and rapid indication of the compressive strength of concrete by means of establishing a suitable correlation between the rebound index and the compressive strength of concrete. In general, the rebound number increases as the strength increases but it is also affected by a number of parameters

It is also pointed out that rebound indices are indicative of compressive strength of concrete to a limited depth from the surface. If the concrete in a particular member has internal micro cracking, flaws or heterogeneity across the cross-section, rebound hammer indices will not indicate the same.

As such, the estimation of strength of concrete by rebound hammer method cannot be held to be very accurate and probable accuracy of prediction of concrete strength in a structure is ± 25 percent. If the relationship between rebound index and compressive strength can be checked by tests on core samples obtained from the structure or standard specimens made with the same concrete materials and mix proportion, then the accuracy of results and confidence there on are greatly increased.



Figure 4.3 Performing Rebound Hammer Test

TABLE 4.4
Hardness criteria for concrete quality grading
(IS 13311 (PART 2):1992)

Average Rebound Number	Quality
Above 40	Very Good Hard Concrete
30 - 40	Good Concrete
20 - 30	Fair Concrete
Below 20	Poor Concrete

- **Co-Relation between CTM and Rebound**

The most suitable method of obtaining the correlation between compressive strength of concrete and rebound number is to test the concrete cubes using compression testing machine as well as using rebound hammer simultaneously. First the rebound number of concrete cube is taken and then the compressive strength is tested on compression testing machine. The fixed load required is of the order of 7 N/ mm^2 when the impact energy of the hammer is about 2.2 Nm. The load should be increased for calibrating rebound hammers of greater impact energy and decreased for calibrating rebound hammers of lesser impact energy. The test specimens should be as large a mass as possible in order to minimize the size effect on the test result of a full scale structure. 150mm cube specimens are preferred for calibrating rebound hammers of lower impact energy (2.2Nm), whereas for rebound hammers of higher impact energy, for example 30 Nm, the test cubes should not be smaller than 300mm.

The concrete cube specimens should be kept at room temperature for about 24 hours after taking it out from the curing pond, before testing it with the rebound hammer. To obtain a correlation between rebound numbers and strength of wet cured and wet tested cubes, it is necessary to establish a correlation between the strength of wet tested cubes and the strength of dry tested cubes on which rebound readings are taken. A direct correlation between rebound numbers on wet cubes and the strength of wet cubes is not

recommended. Only the vertical faces of the cubes as cast should be tested. At least nine readings should be taken on each of the two vertical faces accessible in the compression testing machine when using the rebound hammers. The points of impact on the specimen must not be nearer an edge than 20mm and should be not less than 20mm from each other. The same points must not be impacted more than once.

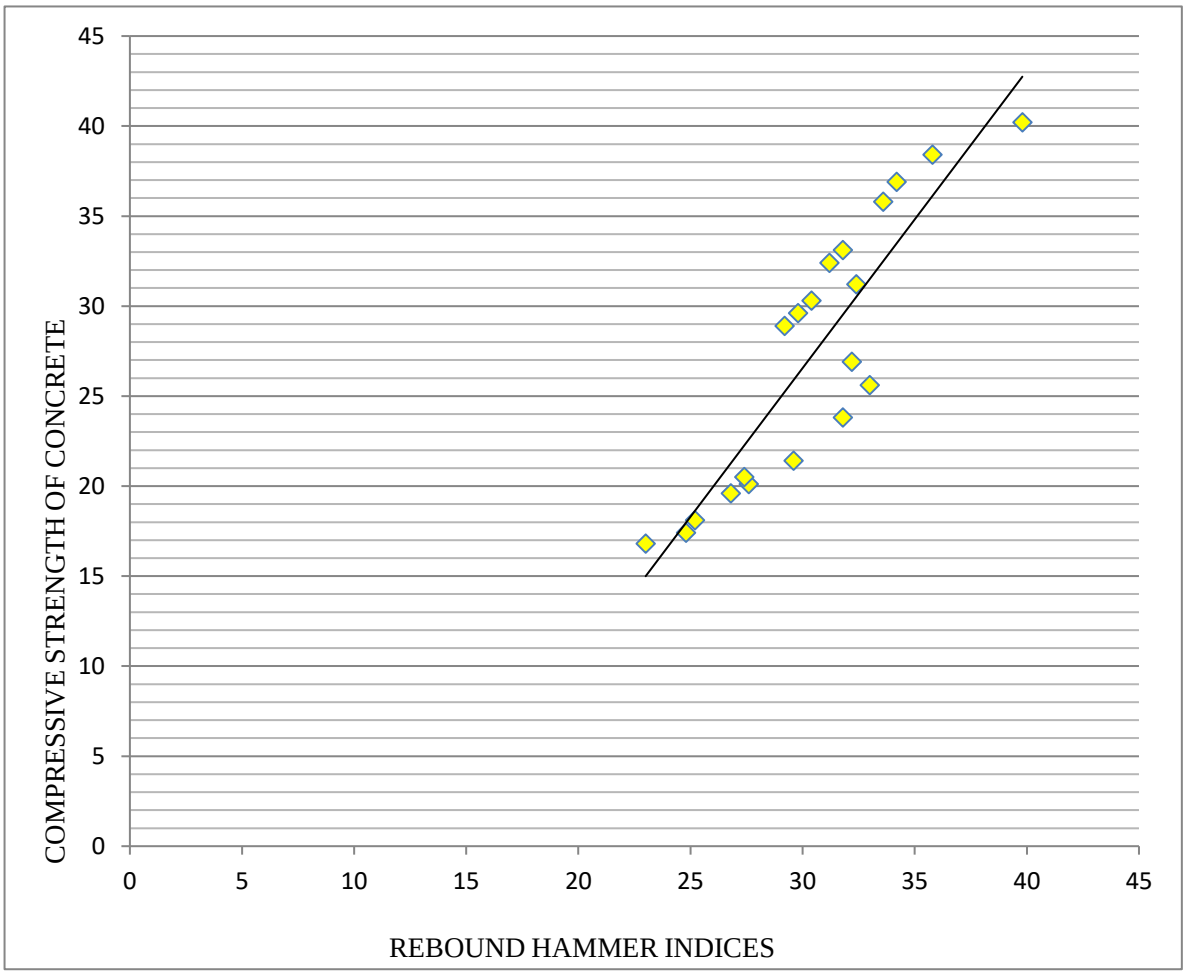
- **Comparison of Results**

Using the rebound values for strength estimation, a correlation coefficient of **0.79** is achieved.

TABLE 4.5ReboundNumbervs Compressive Strength

R	Compressive Strength by CTM
23	16.8
24.8	17.4
25.2	18.1
26.8	19.6
27.6	20.1
27.4	20.5
29.6	21.4
31.8	23.8
33	25.6
32.2	26.9
29.2	28.9
29.8	29.6
30.4	30.3
32.4	31.2

31.2	32.4
31.8	33.1
33.6	35.8
34.2	36.9
35.8	38.4
39.8	40.2



GRAPH 4.1 Compressive Strengthvs Rebound Hammer Indices

y = 1.6516x - 22.991

Accuracy obtained 0.798

4.2.3 Ultrasonic Pulse Velocity

The quality of concrete in terms of uniformity, incidence or absence of internal flaws, cracks and segregation, etc., indicative of the level of workmanship employed; can thus be assessed using the guidelines given in Table 4.6, which have been evolved for characterizing the quality of concrete in structures in terms of the ultrasonic pulse velocity.

Since actual values of the pulse velocity obtained, depend on a number of parameters, any criterion for assessing the quality of concrete on the basis of pulse velocity as given in Table 4.6 can be held as satisfactory only to a general extent. However, when the comparison is made amongst different parts of a structure, which have been built at the same time with supposedly similar materials, construction practices and supervision, the assessment of quality becomes more meaningful and reliable.



Figure 4.4 Performing Ultrasonic Pulse Velocity Test

TABLE 4.6
Velocity criteria for concrete quality grading
(IS 13311 (PART 1):1992)

Pulse Velocity		Quality
km/sec	**ft/sec**	
> 4.5	>15	Excellent
3.5 – 4.5	12 – 15	Good
3.0 – 3.5	10 – 12	Doubtful
2.0 – 3.0	7 – 10	Poor
< 2.0	< 7	Very Poor

- **Co-Relation between CTM and UPV**

The ultrasonic pulse velocity of concrete is mainly related to its densityand modulus of elasticity. This in turn, depends upon the materials and mix proportions used in makingconcrete as well as the method of placing, compaction and curing of concrete. For example, if the concrete is not compacted as thoroughly as possible, or if there is segregation of concrete during placing or there are internal cracks or flaws, the pulse velocity will be lower, although the same materials and mix proportions are used. The quality of concrete in terms of uniformity, incidence or absence of internal flaws, cracks and segregation, etc., indicative of the level of workmanship employed; can thus be assessed using the guidelines given in Table 2, which have been evolved for characterizing the quality of concrete in structures in terms of the ultrasonic pulse velocity.

The assessment of compressive strength of concrete from ultrasonic pulse velocity values is not adequate because the statistical confidence of the correlation between ultrasonic pulse velocity and the compressive strength of concrete is not very high. The reason is that a large number of parameters are involved, which influence the pulse velocity and compressive strength of concrete to different extents. However if actual concrete materials ’ and mix proportions adopted in a particular structure are available, then estimate of

concrete strength can be made by establishing suitable correlation between the pulse velocity and the compressive strength of concrete specimens made with such materials and mix proportions, under environmental conditions similar to that in the structure. The estimated strength may vary from the actual strength by ±20 percent. The correlation so obtained may not be applicable for concrete of another grade or made with different types of material.

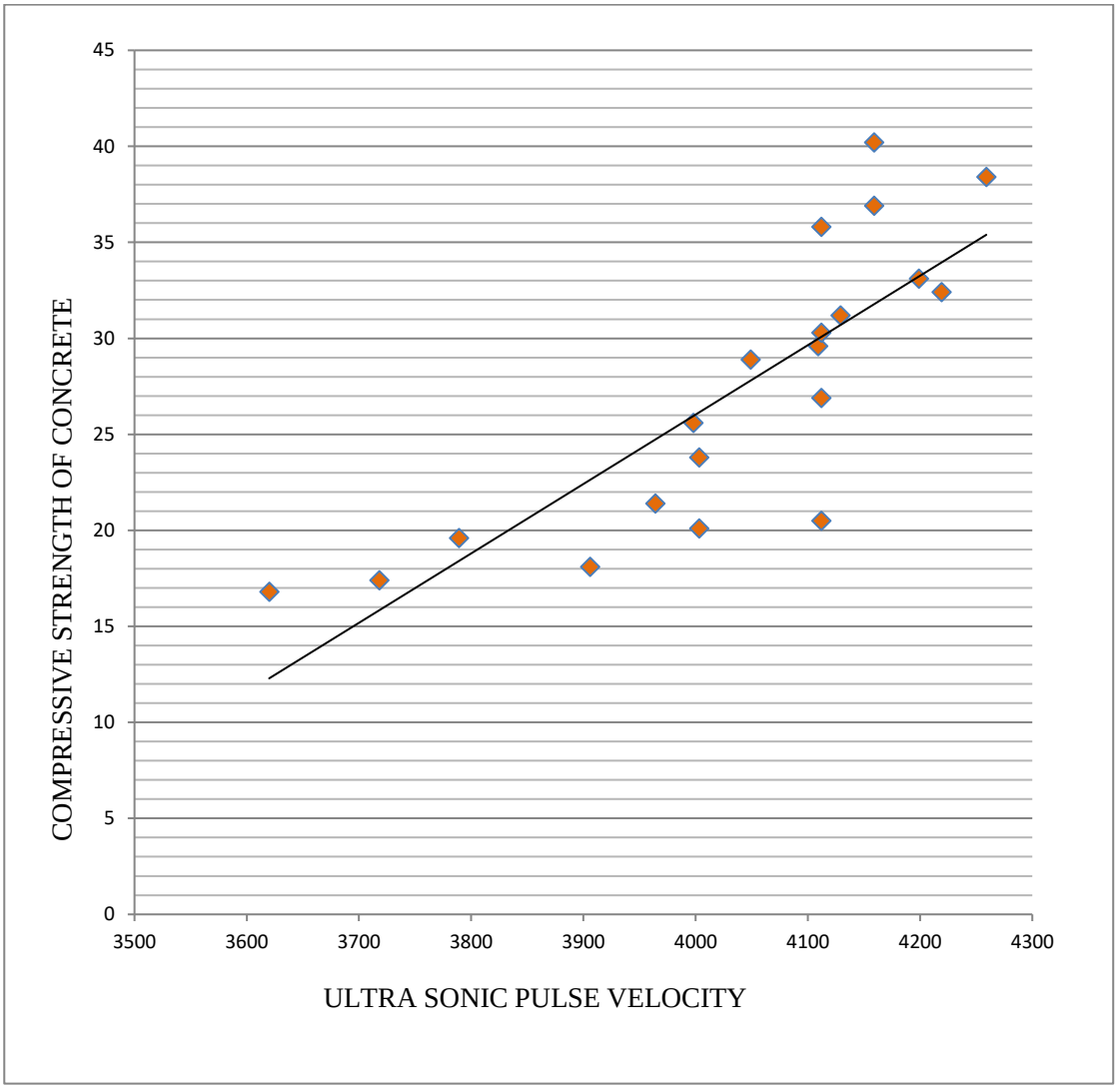
- **Comparison of Results**

Using the ultrasonic pulse velocity for strength estimation, a correlation coefficient of 0.67 isachieved:

TALBE 4.7 UPV vs Compressive Strength

UPV	Compressive Strength by CTM
3620	16.8
3718	17.4
3906	18.1
3789	19.6
4003	20.1
4112	20.5
3964	21.4
4003	23.8
3998	25.6
4112	26.9
4049	28.9
4109	29.6
4112	30.3
4129	31.2

4219	32.4
4199	33.1
4112	35.8
4159	36.9
4259	38.4
4159	40.2



GRAPH 4.2 Compressive StrengthvsUltrasonic Pulse Velocity

y =0.0362x - 118.57

Accuracy obtained 0.6724

4.2.4 SonReb (Combined Interpretation of Results)

In order to create a SONREB curve for estimating the compressive strength of a particular concrete itis necessary to carry out both ultrasonic pulse velocity measurements and rebound measurementson samples of the concrete and then to correlate this to the compressive strength determined in thepress.

TABLE 4.8 Sonreb Analyses

Compressive Strength by CTM	UPV	R
16.8	3620	23
17.4	3718	24.8
18.1	3906	25.2
19.6	3789	26.8
20.1	4003	27.6
20.5	4112	27.4
21.4	3964	29.6
23.8	4003	31.8
25.6	3998	33
26.9	4112	32.2
28.9	4049	29.2
29.6	4109	29.8
30.3	4112	30.4
31.2	4129	32.4
32.4	4219	31.2
33.1	4199	31.8
35.8	4112	33.6

36.9	4159	34.2
38.4	4259	35.8
40.2	4159	39.8

- **The SONREB curve is expressed in the format:**

The format of the curve is:

Compressive Strength,

$$fck = a.\ V^b.\ R^c$$

Where:

a, b and c are constants

V is the ultrasonic pulse velocity in m/s

R is the rebound value.

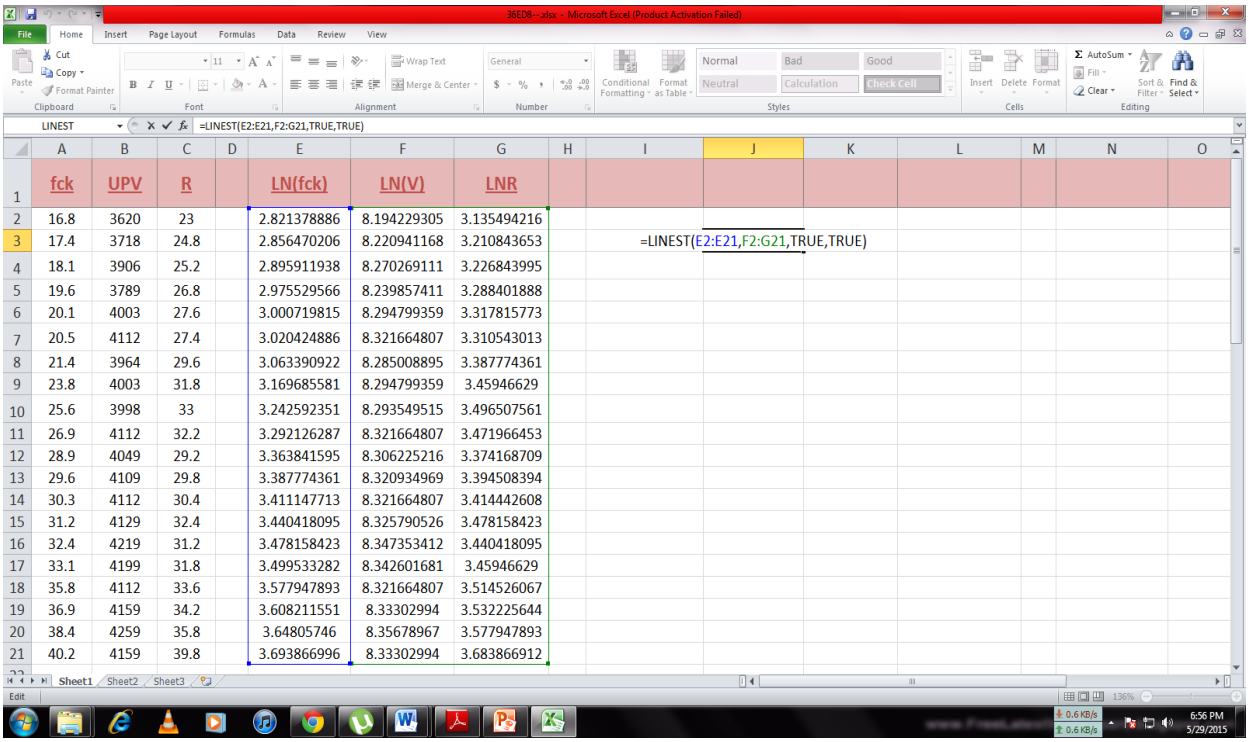
The LINEST function of EXCEL can be used to do this. Below is an extract from the EXCEL help file: “The LINEST function calculates the statistics for a line by using the "least squares" method to calculate a straight line that best fits your data, and then returns an array that describes the line. You can also combine LINEST with other functions to calculate the statistics for other types of models that are linear in the unknown parameters, including polynomial, logarithmic, exponential, and power series. Because this function returns an array of values, it must be entered as an array formula.

In the case of SONREB we have a power series curve so we have to work with the natural logarithmsof the data in order to use LINEST. To do this we first use the function “LN”.

TABLE 4.9 Tables Logarithmic

LN(fck)	LN(V)	LN(R)
2.821378886	8.194229305	3.135494216
2.856470206	8.220941168	3.210843653
2.895911938	8.270269111	3.226843995
2.975529566	8.239857411	3.288401888
3.000719815	8.294799359	3.317815773
3.020424886	8.321664807	3.310543013
3.063390922	8.285008895	3.387774361
3.169685581	8.294799359	3.45946629
3.242592351	8.293549515	3.496507561
3.292126287	8.321664807	3.471966453
3.363841595	8.306225216	3.374168709
3.387774361	8.320934969	3.394508394
3.411147713	8.321664807	3.414442608
3.440418095	8.325790526	3.478158423
3.478158423	8.347353412	3.440418095
3.499533282	8.342601681	3.45946629
3.577947893	8.321664807	3.514526067
3.608211551	8.33302994	3.532225644
3.64805746	8.35678967	3.577947893
3.693866996	8.33302994	3.683866912

To enter the array function select an array of 3 columns x 5 rows and enter the LINEST function in the top left hand corner of the array as shown:-



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	fck	UPV	R		LN(fck)	LN(V)	LNR								
2	16.8	3620	23		2.821378886	8.194229305	3.135494216								
3	17.4	3718	24.8		2.856470206	8.220941168	3.210843653		=LINEST(E2:E21,F2:G21,TRUE,TRUE)						
4	18.1	3906	25.2		2.895911938	8.270269111	3.226843995								
5	19.6	3789	26.8		2.975529566	8.239857411	3.288401888								
6	20.1	4003	27.6		3.000719815	8.294799359	3.317815773								
7	20.5	4112	27.4		3.020424886	8.321664807	3.310543013								
8	21.4	3964	29.6		3.063390922	8.285008895	3.387774361								
9	23.8	4003	31.8		3.169685581	8.294799359	3.45946629								
10	25.6	3998	33		3.242592351	8.293549515	3.496507561								
11	26.9	4112	32.2		3.292126287	8.321664807	3.471966453								
12	28.9	4049	29.2		3.363841595	8.306225216	3.374168709								
13	29.6	4109	29.8		3.387774361	8.320934969	3.394508394								
14	30.3	4112	30.4		3.411147713	8.321664807	3.414442608								
15	31.2	4129	32.4		3.440418095	8.325790526	3.478158423								
16	32.4	4219	31.2		3.478158423	8.347353412	3.440418095								
17	33.1	4199	31.8		3.499533282	8.342601681	3.45946629								
18	35.8	4112	33.6		3.577947893	8.321664807	3.514526067								
19	36.9	4159	34.2		3.608211551	8.33302994	3.532225644								
20	38.4	4259	35.8		3.64805746	8.35678967	3.577947893								
21	40.2	4159	39.8		3.693866996	8.33302994	3.683866912								

Figure 4.5(a) EXCEL Screenshots

The array will show the following data. They key to the data is shown in the table on the right. The coefficients “b” for the pulse velocity and “c” for the rebound hammer can be read directly. For the constant “a” it is necessary to convert it back to the correct form by using an exponential function.

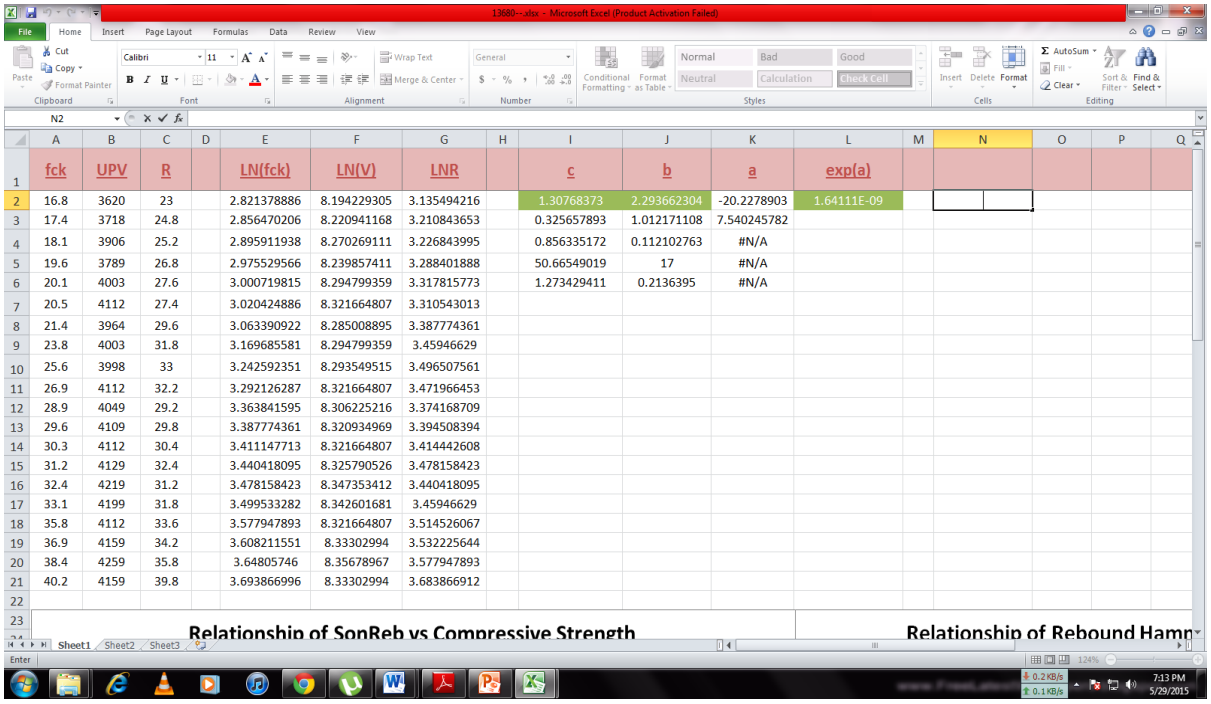
TABLE 4.10 Constants Obtained

c	b	a
1.30768373	2.293662304	-20.2278902
0.325657893	1.012171108	7.540245782
0.856335172	0.112102763	#N/A
50.66549019	17	#N/A
1.273429411	0.2136395	#N/A

Coefficient c = 1.30768373

Coefficient b = 2.29366230

Coefficient a = 1.64111E-09 =EXP(-20.2278902)

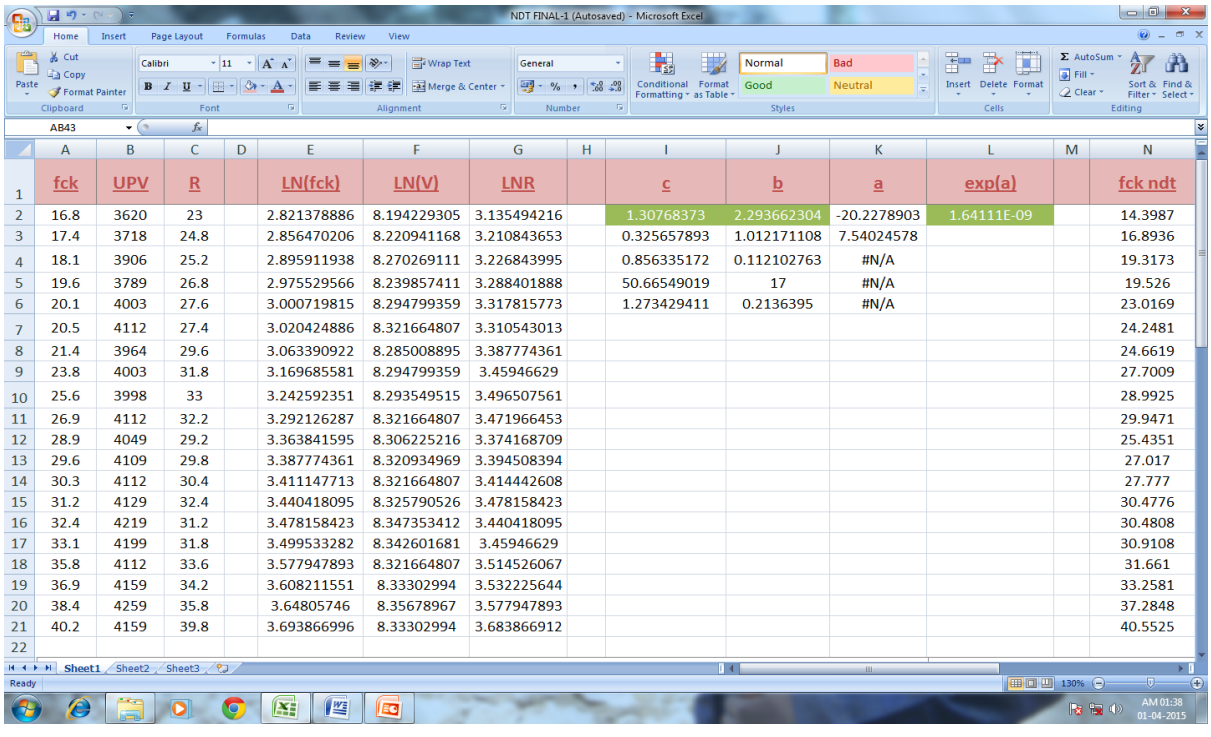


fck	UPV	R	LN(fck)	LN(V)	LNR	c	b	a	exp(a)
16.8	3620	23	2.821378886	8.194229305	3.135494216	1.30768373	2.293662304	-20.2278903	1.64111E-09
17.4	3718	24.8	2.856470206	8.220941168	3.210843653	0.325657893	1.012171108	7.540245782	
18.1	3906	25.2	2.895911938	8.270269111	3.226843995	0.856335172	0.112102763	#N/A	
19.6	3789	26.8	2.975529566	8.239857411	3.288401888	50.66549019	17	#N/A	
20.1	4003	27.6	3.000719815	8.294799359	3.317815773	1.273429411	0.2136395	#N/A	
20.5	4112	27.4	3.020424886	8.321664807	3.310543013				
21.4	3964	29.6	3.063390922	8.285008895	3.387774361				
23.8	4003	31.8	3.169685581	8.294799359	3.45946629				
25.6	3998	33	3.242592351	8.293549515	3.496507561				
26.9	4112	32.2	3.292126287	8.321664807	3.471966453				
28.9	4049	29.2	3.363841595	8.306225216	3.374168709				
29.6	4109	29.8	3.387774361	8.320934969	3.394508394				
30.3	4112	30.4	3.411147713	8.321664807	3.414442608				
31.2	4129	32.4	3.440418095	8.325790526	3.478158423				
32.4	4219	31.2	3.478158423	8.347353412	3.440418095				
33.1	4199	31.8	3.499533282	8.342601681	3.45946629				
35.8	4112	33.6	3.577947893	8.321664807	3.514526067				
36.9	4159	34.2	3.608211551	8.33302994	3.532225644				
38.4	4259	35.8	3.64805746	8.35678967	3.577947893				
40.2	4159	39.8	3.693866996	8.33302994	3.683866912				

Figure 4.5(b) EXCEL Screenshots

So the SONREB curve for this set of data would be:

$$f_{ck} = (1.64111E\text{-}09) \cdot V^{2.293662304} \cdot R^{1.30768373}$$



fck	UPV	R	LN(fck)	LN(V)	LNR	c	b	a	exp(a)	fck ndt
16.8	3620	23	2.821378886	8.194229305	3.135494216	1.30768373	2.293662304	-20.2278903	1.64111E-09	14.3987
17.4	3718	24.8	2.856470206	8.220941168	3.210843653	0.325657893	1.012171108	7.54024578		16.8936
18.1	3906	25.2	2.895911938	8.270269111	3.226843995	0.856335172	0.112102763	#N/A		19.3173
19.6	3789	26.8	2.975529566	8.239857411	3.288401888	50.66549019	17	#N/A		19.526
20.1	4003	27.6	3.000719815	8.294799359	3.317815773	1.273429411	0.2136395	#N/A		23.0169
20.5	4112	27.4	3.020424886	8.321664807	3.310543013					24.2481
21.4	3964	29.6	3.063390922	8.285008895	3.387774361					24.6619
23.8	4003	31.8	3.169685581	8.294799359	3.45946629					27.7009
25.6	3998	33	3.242592351	8.293549515	3.496507561					28.9925
26.9	4112	32.2	3.292126287	8.321664807	3.471966453					29.9471
28.9	4049	29.2	3.363841595	8.306225216	3.374168709					25.4351
29.6	4109	29.8	3.387774361	8.320934969	3.394508394					27.017
30.3	4112	30.4	3.411147713	8.321664807	3.414442608					27.777
31.2	4129	32.4	3.440418095	8.325790526	3.478158423					30.4776
32.4	4219	31.2	3.478158423	8.347353412	3.440418095					30.4808
33.1	4199	31.8	3.499533282	8.342601681	3.45946629					30.9108
35.8	4112	33.6	3.577947893	8.321664807	3.514526067					31.661
36.9	4159	34.2	3.608211551	8.33302994	3.532225644					33.2581
38.4	4259	35.8	3.64805746	8.35678967	3.577947893					37.2848
40.2	4159	39.8	3.693866996	8.33302994	3.683866912					40.5525

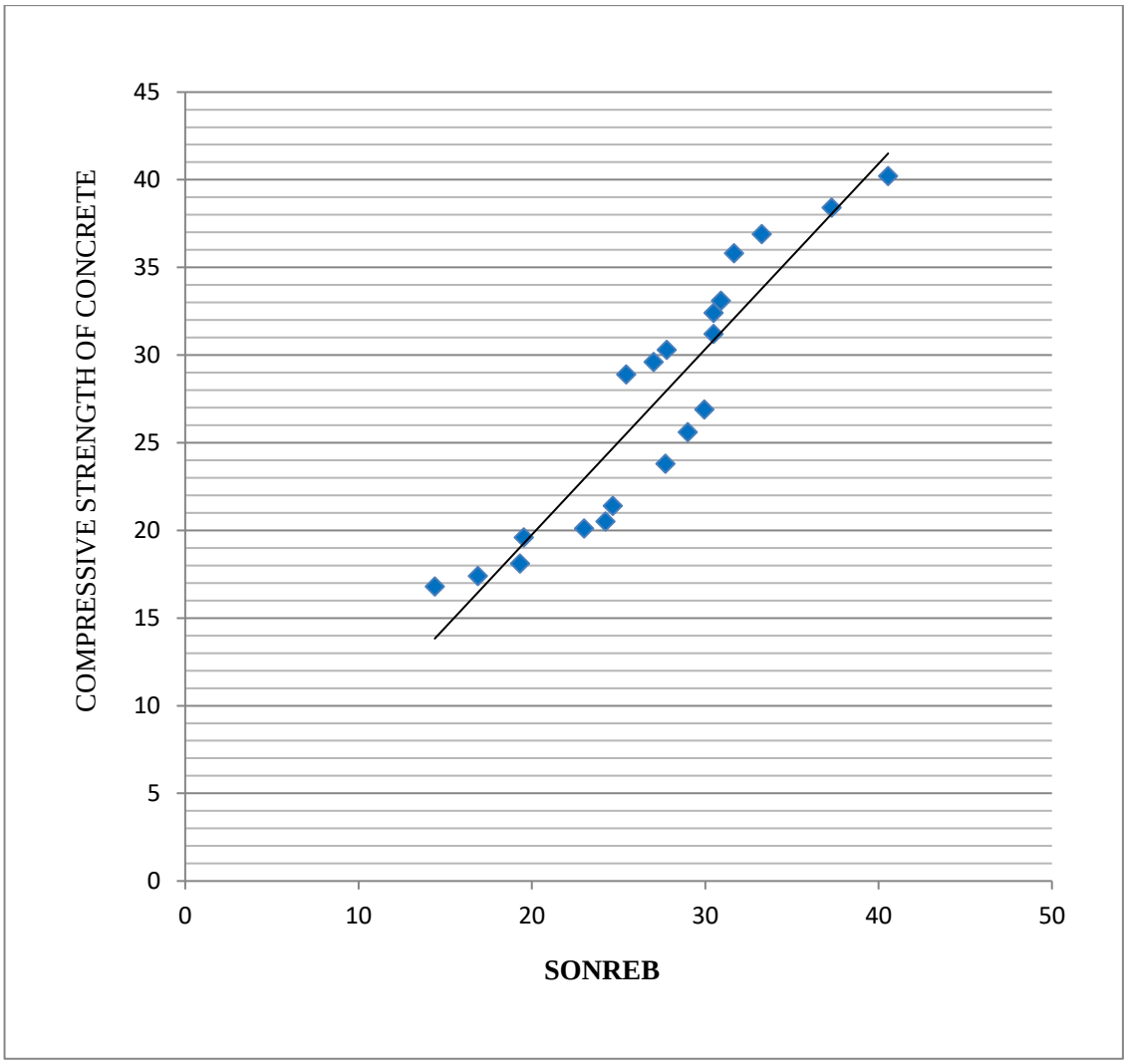
Figure 4.5(c) EXCEL Screenshots

- **Comparison of Results**

Comparison of CTM vs. SonReb

TABLE 4.11 Compressive Strength (CTM) vs. Compressive Strength(SONREB)

Fck CTM	Fck SonReb (NDT)
16.8	14.3987
17.4	16.8936
18.1	19.3173
19.6	19.526
20.1	23.0169
20.5	24.2481
21.4	24.6619
23.8	27.7009
25.6	28.9925
26.9	29.9471
28.9	25.4351
29.6	27.017
30.3	27.777
31.2	30.4776
32.4	30.4808
33.1	30.9108
35.8	31.661
36.9	33.2581
38.4	37.2848
40.2	40.5525



GRAPH 4.3 Compressive Strength(CTM) vs Compressive Strength(SONREB)

y = 1.058x - 1.4048

Accuracy Obtained 0.8669

In comparison, combining the two methods by using the SONREB data gives a significantly higher correlation coefficient of 0.86 as we have seen.

- **PUNDIT LAB +**

Pundit Lab+ allows compressive strength estimates to be made using pulse velocity measurements or a combination of pulse velocity and rebound hammer measurements.

To do this it is necessary to create a conversion curve and upload this onto the instrument.

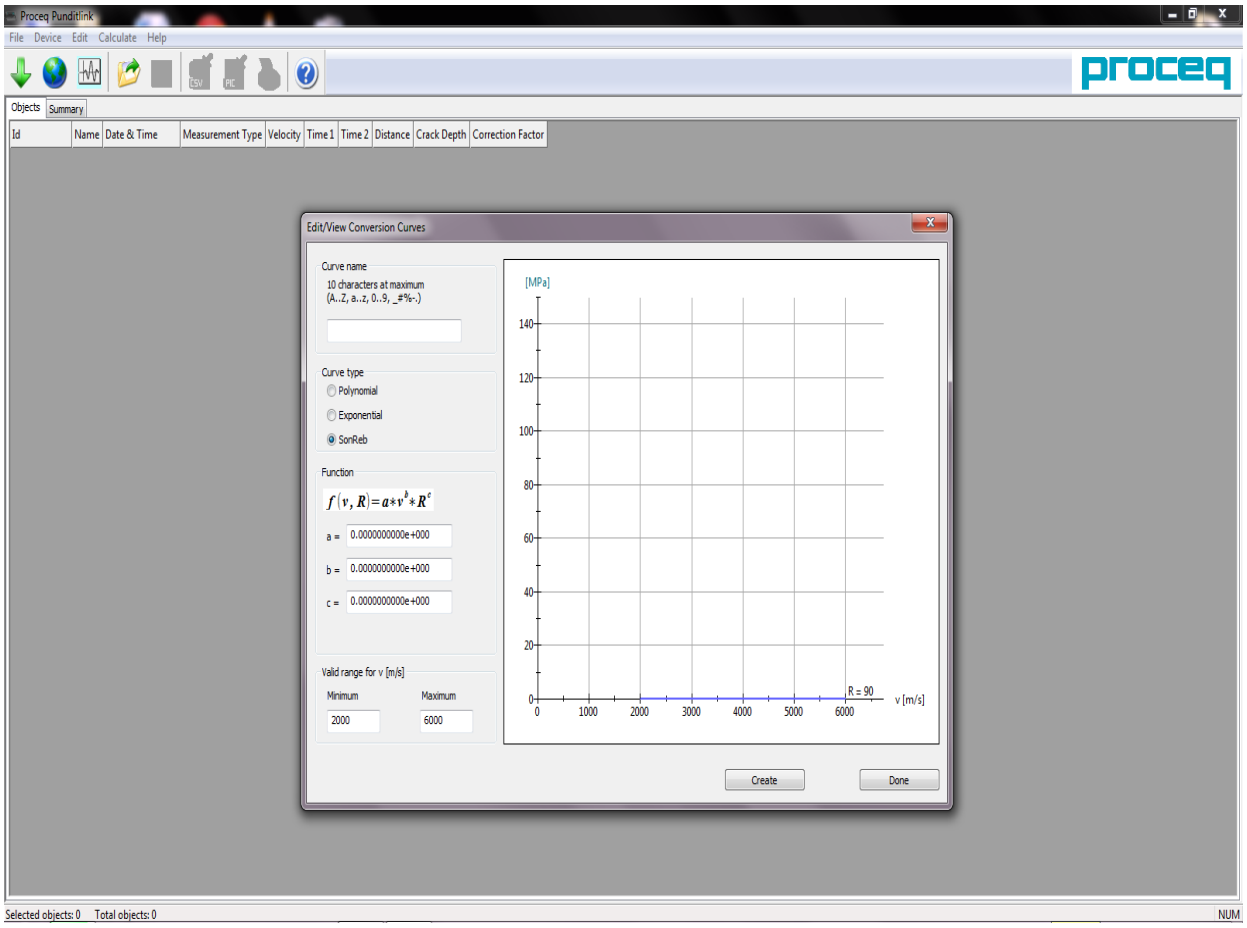


Figure 4.6 PUNDIT LAB+ SOFTWARE Screenshot

Pundit Lab+ allows either polynomial curves or exponential curves to be programmed and in the case of a combined ultrasonic/rebound value measurement a curve based on the SONREB (SONic +REBound) method may be entered.

Pundit Lab+ only.Estimating compressive strength using pulse velocity alone or in combination with a rebound hammer.

4.2.5 Costing of NDT And Its Economy

The use of nondestructive technique over the destructive technique is that, in **destructive testing**, or (Destructive Physical Analysis DPA) tests are carried out to the specimen's failure, in order to understand a specimen's structural performance or material behavior under different loads. But in case of Nondestructive testing (NDT) the specimen is not physically damaged but is tested in its original state, so that no damage is caused to the original structure and the strength is calculated. This also saves the money required for extracting the specimen out of a structure for Destructive testing

Testing by Nondestructive technique is comparatively economical than the Destructive testing. As in the destructive technique the cost of testing a single specimen is too high when compared to NDT.

- **Cost of Destructive testing**.

  Per inch of core extraction cost: - Rs250/-(APPROX.)

  Standard core specimen size to be extracted: - 6 inch (minimum)

  Therefore core extraction cost: - Rs 1500/-

  Testing of core specimen on compression testing machine CTM:- Rs500/-(APPROX.)

  Therefore overall cost of Destructive Technique:- **Rs 2000/- (APPROX.)**

- **Cost of Non Destructive technique.**

  Testing of per point of specimen by REBOUND HAMMER: - Rs 250/-

  Testing per point of specimen by Ultrasonic Pulse Velocity: - Rs 250/-

  Therefore overall cost of testing a specimen: - **Rs500/-**

As we can see that testing a specimen by NDT is comparatively cheaper than testing a specimen by Destructive Technique, so NDT can be preferred over the Destructive Technique when the economy of a certain project is to be focused.

CHAPTER - 5

CONCLUSION

5.1 Conclusions

The SonReb method is advantageous because the UPV and RN have different sensitivities to theparameters that are important when predictingf_{ck}.In general, most of the models used to study the on-site concrete compressive strength consist ofmathematical rules and expressions that try to capture relationship between NDT parameters.Generally these mathematical models based on experimental data are presented in regressionforms.In this study performances of the most used formulations in Italy for the estimation of thecompressive strength of concrete utilizing NDT are compared.

The results obtained from the study indicate the good estimation potential of the proposedformulations in the evaluation of concrete compressive strength, starting only from the nondestructive parameters.In fact, the root mean squared error (*RMSE*) values obtained for all the cases are to be consideredreasonably low in relation to those obtained with the other formulas in the literature, thusindicating the accuracy of the proposed estimations.

This method works when using a single NDT, but is improved considerably when combining the information provided by two NDT measurements which shows, as much as possible, adverse sensitivity to the same uncontrolled parameter. For instance, if the main uncontrolled parameter is water content in the concrete, UPV and Rebound Hammer are good candidates for being the two best techniques to use. In other cases (Influence of chlorides, of carbonation, etc.) other techniques like radar measurements or electric resistivity, can be possible candidates.

1. The comparison carried out shows how the use of known destructive methodologies (cores), associated with a non-destructive method (SonReb) allows to obtain a higher level of knowledge and a greater accuracy on the estimation of the concrete compressive strength if the relationship is calibrated on the individual building

2. The SonReb method provides a reliable assessment of the onsite concrete compressive strength which allows obtaining the required levels of knowledge.
3. This, in turn, allows limiting the number of destructive tests needed to properly characterize concrete strength in existing buildings.
4. Simply using the rebound values alone for strength estimation, a correlation coefficient of 0.798 is achieved
5. Using the ultrasonic pulse velocity for strength estimation, a correlation coefficient of 0.672 is achieved.
6. Combining the two methods by using the SONREB higher correlation coefficient of 0.867 as we have seen.
7. From the correlation generated, we can find more accurate compressive strength than the previous.

Just by knowing the Rebound Hammer Number and UPV, strength can be calculated

5.2 Scope for NDT

Depending upon the possible solution we have found, we checked for the solution feasible for the convenience of use. Here we are summering all the possible solution observations results and conclusion.

- The main scope of NDT application is to help ensure the safety, integrity and reliability of plant items such as pressure vessels, boiler,heat exchangers, and pipe work pipeline.
- The inspection of concrete structure is a part of plant life assessment since these materials often forms an important part of the construction of such plant item. The failure of concrete structure presents arisk in itself and may contribute to the consequent failure of other plant items.
- The present work focuses on the study of reliability in interpreting non-destructive testing result of concrete structure.it will be conducted on normal hardened concrete ranging from 20 to 55 MPa. And in laboratory the study will be restricted to the following properties: compressive strength, uniformity of concrete and covercrete (concrete cover).
- In aircraft manufacture and maintenance NDT is used.

APPENDIX A














M20
3/2/2015
M45
4/2/2015
M50
4/2/2015
M60
4/2/2015
M35
3/2/2015
M40
3/2/2015
M30
3/2/2015
M35
4/2/2015











APPENDIX B

THE USE OF COMBINED NON DESTRUCTIVE TESTING IN THE CONCRETE STRENGHT ASSESSMENT FROM LABORATORY SPECIMENS AND EXISTING BUILDINGS

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Abstract: -Non Destructive Testing (NDT) as the name implies refers to a test that does not impair the intended performance of the element, member or structure under investigation.In this paper a series of non-destructive tests has been performed with the purpose to investigate on the mechanical properties of the concrete employed in the civil buildings.

A series of specimens were prepared in order to correlate the "in situ" concrete strengths obtained by combined non-destructive method with the cubical strength obtained by destructive methods. The combined method *(SonReb Method)* was used to quality control and strength estimation of the concrete.This combined method requires short time to obtain the results, it's a non-invasive method and it does not affect the resistance of structural elements.

Finally the investigation and comparison of experimental results of non-destructive tests and cubical strength with the help of statistical data obtained by testing of specimen as per recommended procedures by IS 13311:1992 and IS 516:1959 respectively.

From this exhaustive and extensive experimental work it was found that SonReb method, combining Schmidt rebound hammer and Ultrasonic Pulse Velocity methods, allows compensating the limits and the uncertainty typical of each method. The use of the combined methods (SonReb) increases the accuracy of the estimation of the in situ concrete compressive strength.

The research aims on the rehabilitation of reinforced concrete buildings that begin to show signs of decay and deterioration.

Keywords: - Concrete, Concrete compressive strength, Non-destructive investigations, Combined Methods, SonReb Methods.

I. INTRODUCTION

Concrete has been most widely used as construction material for over 100 years, because it is strong in compression. The evaluation of the concrete compressive strength is a fundamental step for assessment of existing reinforced concrete building according to the last seismic codes. This valuation can be conducted by the use of both destructive and non-destructive method. Through the concrete compressive strength, other concrete

properties like elastic modulus, tensile strength and durability can be obtained.

Service life of concrete is found to be limiting in various environmental degrading factors as it is exposed to it. This therefore has brought about the need for test method to measure the in-place properties of concrete for quality assurance and for evaluation of existing conditions. Since such test are expected as non-impairing the function of the structure and allow for re-testing at the same location to evaluate the changes in property at some other point in time, these methods should be non-destructive.

The combination of several techniques of non-destructive testing is often used empirically, combining two techniques mostly used to enhance the reliability of the estimate compressive strength of concrete; the principle is based on correlations between observed measurements and the desired property.The standardized combine method and the most widely used internationally is SonReb method. First born and established in Romania then developed in Australia and in Europe.The best approach is generally to develop a relationship of correlation between the Ultra Sonic Pulse Velocity, the index of rebound hammer and the compressive strength of standardized laboratory specimen.

II. TEST SPECIMEN

A. Materials used

[1] *Cement:* Ordinary Portland cement (OPC) of 53 grade of Birla Super Cement is used in this experimental work. Weight of each cement bag is 50 kg.

[2] *Fine aggregates:* Crushed sand having specific gravity 2.77gm/cc and Fineness modulus as 3.15

[3] *Coarse aggregate:* Consist of 10 mm and 20 mm crushed aggregate. 10 mm aggregate having specific gravity 2.91gm/cc and FM as 2.012. 20mm aggregate having specific gravity 2.88gm/cc and FM as 2.1.

B. Mix proportion and casting procedure

Hand mixing over a mixing tray was done throughout. Coarse aggregates were placed first in the tray followed by crushed sand, and then cement. The materials were dry mixed thoroughly for 1 min.before adding water.Mixing continued for further few minutes after adding water.Concrete was then placed in IS specified moulds in three layers, each layer was being compacted by standard tamping road with more than 35 strokes.Exposed surface was finished with trowel to avoid uneven surface.

A total of 60 concrete specimens 150x150x150mm was designed and fabricated.Specimens were prepared to obtain characteristic cube strength of 15 MPa, 20 MPa, 25 MPa, 30 MPa, 35 MPa and 40 MPa. In particular, 10 specimens of each grade. Specimens were cured by immersing them in curing tank for 28 days.

TABLE 1
MIX PROPORTION AND CONCRETE BATCH

Grade	Cement (kg/m^3)	Aggregate (kg/m^3)		Water (lit/m^3)
		Fine	Coarse	
M15	270	711	1460	135
M20	320	794	1138	176
M25	340	775	1112	185
M30	380	760	1090	187
M35	410	735	1053	200
M40	430	718	1030	185

III. TEST PROGRAM

A. Rebound Hammer test on cubes

The Schmidt rebound hammer is principally a surface hardness tester, which works on the principle that the rebound of an elastic mass depends on the hardness of the surface against which the mass impinges. There is seemingly theoretical relationship between the strength of concrete and the rebound number of the hammer.

The weight of the Schmidt rebound hammer is about 1.8 kg and is suitable for both laboratory and field purpose. The rebound distance of the hammer mass is measured on an arbitrary scale ranging from 10 to 100. The rebound distance of the hammer is recorded as a "rebound number" corresponding to the position of the rider on the scale.

TABLE 2
HARDNESS CRITERIA FOR CONCRETE QUALITY GRADING (IS 13311 (PART 2):1992)

Average Rebound Number	Quality
Above 40	Very Good Hard Concrete
30 - 40	Good Concrete
20 - 30	Fair Concrete
Below 20	Poor Concrete

B. Ultrasonic Pulse Velocity test on cubes

The equipment consists basically of an electrical pulse generator, a pair of transducers, an amplifier and an electronic timing device for measuring the time interval between the initiation of a pulse generated at the transmitting transducer and its arrival at the receiving transducer.

Pulse velocity (in km/s or m/s) is given by:

$$v = \frac{L}{T}$$

where,

v is the longitudinal pulse velocity,
L is the path length,
T is the time taken by the pulse to travel path length.

TABLE 3
CONCRETE QUALITY GRADING USING VELOCITY CRITERION (IS 13311 (PART 1):1992)

Pulse Velocity		Quality
km/sec	ft/sec	
> 4.5	> 15	Excellent
3.5 – 4.5	12 – 15	Good
3.0 – 3.5	10 – 12	Doubtful
2.0 – 3.0	7 – 10	Poor
< 2.0	< 7	Very Poor

C. Compression testing of cubes

Compression testing machine of capacity 2000 kN is used for compression testing of cube as casted of size 150 x 150 x 150 mm and capable of giving load at the rate of 140 kg/sq.cm/min. Testing of the concrete cubes is tested under CTM at the age of 28 days. The wet cubes were placed in the machine between wiped and cleaned loading surfaces and load is given approximately at the rate of 140 kg/sq.cm/min. and ultimate crushing load is noted to calculate crushing strength of concrete according to IS: 516-1959.

The measuring strength of specimen is calculated by dividing the maximum load applied to the specimen during the test by the cross section area.

D. Calculation

The compressive strength of the specimen shall be expressed as f_{ck}.

In the case of the SonReb method the law of correlation among compressive strength, Rebound Hammer index and ultrasonic velocity can be expressed as (Pucinotti 2005):

$$f_{ck} = a \cdot V^b \cdot R^c \qquad \ldots (i)$$

where,

a, b, c are constants
V - Ultrasonic Pulse Velocity
R - Rebound Number

IV. RESULT AND DISCUSSION

Among the obtained 60 results of Rebound Hammer, Ultrasonic Pulse Velocity and Compression testing, 20 random results are selected to obtain a, b and c constants.

TABLE 4
RESULTS

R	UPV	Compressive Strength by CTM
23	3620	16.8
24.8	3718	17.4
25.2	3906	18.1
26.8	3789	19.6
27.6	4003	20.1
27.4	4112	20.5
29.6	3964	21.4
31.8	4003	23.8
33	3998	25.6
32.2	4112	26.9
29.2	4049	28.9
29.8	4109	29.6
30.4	4112	30.3
32.4	4129	31.2
31.2	4219	32.4
31.8	4199	33.1
33.6	4112	35.8
34.2	4159	36.9
35.8	4259	38.4
39.8	4159	40.2

So the question is how to determine the constants *a, b* and *c* from the data.The LINEST function of EXCEL can be used to do this.

The *a, b* and *c* obtained are:

a = 1.64111E-09
b = 2.293662304
c = 1.30768373

So the law of correlation among compressive strength, Rebound Hammer index and ultrasonic velocity for this set of data would be:

$$f_{ck} = (1.64111E\text{-}09) \cdot V^{2.293662304} \cdot R^{1.30768373} \quad \ldots (ii)$$

TABLE 5
COMPRESSIVE STRENGTH BY SONREB USING FORMULA (*ii*)

f_{ck} (SONREB)
14.3987
16.8936
19.3173
19.526
23.0169
24.2481
24.6619
27.7009
28.9925
29.9471
25.4351
27.017
27.777
30.4776
30.4808
30.9108
31.661
33.2581
37.2848
40.5525

By using the above data, we can also see the effectiveness of the SonReb method.

i. Simply using the rebound values alone for strength estimation, a correlation coefficient of 0.798 is achieved.

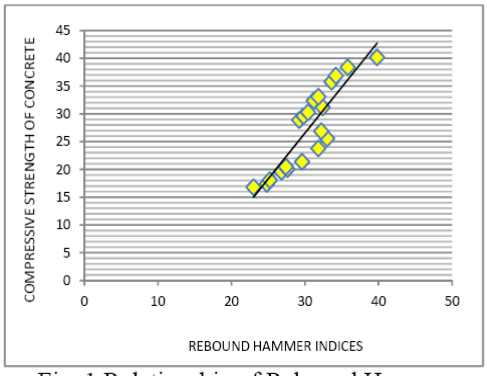


Fig. 1 Relationship of Rebound Hammer vs Compressive Strength

y = 1.6516x - 22.991
Accuracy obtained 0.798

ii. When using the ultrasonic pulse velocity for strength estimation, a correlation coefficient of 0.672 is achieved.

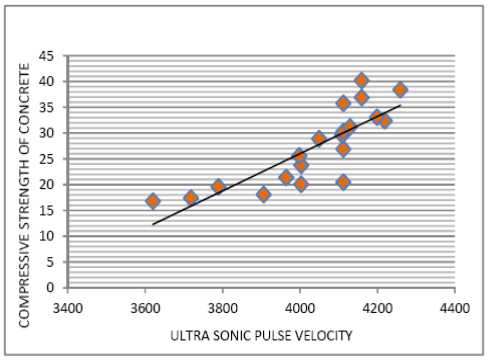


Fig.2 Relationship of Ultrasonic Pulse Velocity vs Compressive Strength

y =0.0362x - 118.57
Accuracy obtained 0.6724

iii. In comparison, combining the two methods by using the SONREB data gives a significantly highercorrelation coefficient of 0.867 as we have seen.

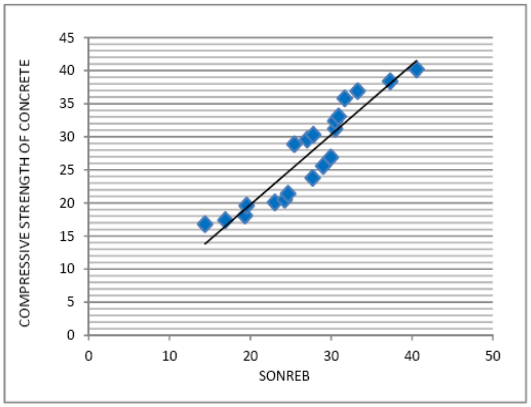


Fig.3 Relationship of SonReb vs Compressive Strength

$$y = 1.058x - 1.4048$$

Accuracy obtained 0.8669

From graph we can say that the regression equation of the Sonreb has a greater accuracy than the previous two equations.

V. CONCLUSION

[1] The comparison carried out shows how the use of known destructive methodologies (cores), associated with a non-destructive method (SonReb) allows to obtain a higher level of knowledge and a greater accuracy on the estimation of the concrete compressive strength if the relationship is calibrated on the individual building

[2] The SonReb method provides a reliable assessment of the onsite concrete compressive strength which allows obtaining the required levels of knowledge.

[3] This, in turn, allows limiting the number of destructive tests needed to properly characterize concrete strength in existing buildings.

[4] Simply using the rebound values alone for strength estimation, a correlation coefficient of 0.798 is achieved

[5] Using the ultrasonic pulse velocity for strength estimation, a correlation coefficient of 0.672 is achieved.

[6] Combining the two methods by using the SONREB higher correlation coefficient of 0.867 as we have seen.

[7] From the correlation generated, we can find more accurate compressive strength than the previous.

[8] Just by knowing the Rebound Hammer Number and UPV, strength can be calculated.

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[2] M. T. Cristofaro, A. D'Ambrisi, M. De Stefano and M. Tanganelli, Mechanical Characterization of Concrete from Existing Buildings with SonReb Method.

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[7] Dallshad K.H. Bzeni, Muhammad A. Ihsan, Estimating Strength of SCC Using Non-Destructive Combined Method

[8] Malhotra V. M., Carino N. J., CRC Handbook on Nondestructive Testing of Concrete, CRC Press, 1991.

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APPENDIX C

- **Certification**

1. At state level Project and Exhibition Competition **INNOVATION–2015**, ALARD COLLEGE OF ENGINEERING AND MANAGEMENT, Pune
    - Nikhil V. Mulik
    - Minal R. Balki
    - Deep S. Chhabria
    - Vijay D. Ghare
    - Vishal S. Tele
    - Nilima S. Avhad
2. **5th International Conference on Biotechnology, Civil and Chemical Engineering (ICBCCE-2015),** Nagpur.
    - Presented a paper entitled **"THE USE OF COMBINED NON DESTRUCTIVE TESTING IN THE CONCRETE STRENGTH ASSESSMENT FROM LABORATORY SPECIMENS AND EXISTING BUILDINGS"**
        - Nikhil V. Mulik
        - Minal R. Balki
        - Deep S. Chhabria
        - Vijay D. Ghare
        - Vishal S. Tele

REFERENCES

1. APPLICATION OF THE COMBINED METHOD FOR EVALUATING THE COMPRESSIVE STRENGTH OF CONCRETE ON SITE
   Samia Hannachi, Mohamed Nacer Guetteche

2. ULTRASONIC PULSE VELOCITY AND REBOUND HAMMER AS NDT TOOLS FOR STRUCTURAL HEALTH MONITORING
   Suresh Chandra Pattanaik
   Research Scholar, School Of Civil Engineering, KIIT University, Bhubaneswar-751024 and Team Head, Dr. Fixit Institute Of Structural Protection & Rehabilitation,Mumbai-400059.

3. MECHANICAL CHARACTERIZATION OF CONCRETE FROM EXISTING BUILDINGS WITH SONREB METHOD
   M.T. Cristofaro, A. D'Ambrisi, M. De Stefano & M. Tanganelli
   Department Of Constructions and Restoration, University Of Florence, Italy R. Pucinotti

4. A COMPARATIVE ANALYSIS OF THE REBOUND HAMMER AND ULTRASONIC PULSE VELOCITY IN TESTING CONCRETE
   J.C. Agunwambaa, T. Adagbab

5. STATISTICAL CASE STUDY ON SCHMIDT HAMMER, ULTRASONIC AND CORE COMPRESSION STRENGTH TESTS' RESULTS PERFORMED ON CORES OBTAINED FROM BEHBAHAN CEMENT FACTORY IN IRAN
   Mohammadreza MAHMOUDIPOUR
   Geotechnical Department ,SANO Consulting Engineers, No.11, Tavanir St.,Valiasr St., Tehran; Iran

6. ESTIMATING STRENGTH OF SCC USING NON-DESTRUCTIVE COMBINED METHOD
   Dallshad K.H. Bzeni1, Muhammed A. Ihsan2

7. USE OF ULTRASOUND TO ESTIMATE DEPTH OF SURFACE OPENING CRACKS IN CONCRETE STRUCTURES
   Roberto C. A. PINTO 1, Arthur MEDEIROS 1, Ivo J. PADARATZ 1, Patrícia B. ANDRADE Civil Engineering Department, Federal University Of Santa Catarina; Florianopolis; Brazil

8. IS CODE 456 (2000): PLAIN AND REINFORCED CONCRETE.

9. IS CODE10262 (2009): GUILINES FOR CONCRETE MIX DESIGN.

10. IS CODE13311 (Part1):1992 NON DESTRUCTIVE TESTING OF CONCRETE- METHODS OF TEST, ULTRASONIC PULSE VELOCITY.

11. IS CODE13311 (Part2):1992 NON DESTRUCTIVE TESTING OF CONCRETE- METHODS OF TEST, REBOUND HAMMER.

12. IS CODE 516 (1959): METHOD OF TEST FOR STRENGTH OF CONCRETE.

13. IS CODE 1199 (1959) METHOD OF SAMPLING AND ANALYSIS OF CONCRETE.

14. IS CODE 383 (1970): SPECIFICATION FOR COARSE AND FINE COARSE AGGREGATES.

15. IS CODE 4031 (1968): FINESS OF CEMENT FOR CONCRETE TECHNOLOGY.